

PSI

Center for Scientific Computing,
Theory and Data

Koopmans functionals in practice

minimisation, screening coefficients, automation, and more...

Edward Linscott

ICTP–MARVEL College, 27 May 2026

$$E_{\text{Koopmans}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(\underbrace{- \int_0^{f_i} \varepsilon_i(f) \, df}_{\text{removes curvature}} + \underbrace{f_i \eta_i}_{\text{restores linearity}} \right)$$

I. Dabo *et al.* *Phys. Rev. B* **82**, 115121 (2010), G. Borghi *et al.* *Phys. Rev. B* **90**, 75135 (2014), N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

$$E_{\text{Koopmans}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(\underbrace{- \int_0^{f_i} \varepsilon_i(f) \, df}_{\text{removes curvature}} + \underbrace{f_i \eta_i}_{\text{restores linearity}} \right)$$

Differences to semi-local functionals:

I. Dabo *et al.* *Phys. Rev. B* **82**, 115121 (2010), G. Borghi *et al.* *Phys. Rev. B* **90**, 75135 (2014), N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

$$E_{\text{Koopmans}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(\underbrace{- \int_0^{f_i} \varepsilon_i(f) \, df}_{\text{removes curvature}} + \underbrace{f_i \eta_i}_{\text{restores linearity}} \right)$$

Differences to semi-local functionals:

- different flavours

I. Dabo *et al.* *Phys. Rev. B* **82**, 115121 (2010), G. Borghi *et al.* *Phys. Rev. B* **90**, 75135 (2014), N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

$$E_{\text{Koopmans}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(\underbrace{- \int_0^{f_i} \varepsilon_i(f) \, df}_{\text{removes curvature}} + \underbrace{f_i \eta_i}_{\text{restores linearity}} \right)$$

Differences to semi-local functionals:

- different flavours
- orbital-density dependence

I. Dabo *et al.* *Phys. Rev. B* **82**, 115121 (2010), G. Borghi *et al.* *Phys. Rev. B* **90**, 75135 (2014), N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

$$E_{\text{Koopmans}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(\underbrace{- \int_0^{f_i} \varepsilon_i(f) \, df}_{\text{removes curvature}} + \underbrace{f_i \eta_i}_{\text{restores linearity}} \right)$$

Differences to semi-local functionals:

- different flavours
- orbital-density dependence
- screening

I. Dabo *et al.* *Phys. Rev. B* **82**, 115121 (2010), G. Borghi *et al.* *Phys. Rev. B* **90**, 75135 (2014), N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

Flavours

Flavours of Koopmans functionals



$$E_{\text{Koopmans}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \eta_i \right)$$

One degree of freedom: what should be the gradient of this linear term?

Flavours of Koopmans functionals



$$E_{\text{KI}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \int_0^1 \varepsilon_i(f) \, df \right)$$

One degree of freedom: what should be the gradient of this linear term?

Flavours of Koopmans functionals



$$E_{\text{KIPZ}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \left\{ \int_0^1 \varepsilon_i(f) \, df - E_{\text{Hxc}}[n_i] \right\} \right)$$

One degree of freedom: what should be the gradient of this linear term?

$$E_{\text{KIPZ}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \left\{ \int_0^1 \varepsilon_i(f) \, df - E_{\text{Hxc}}[n_i] \right\} \right)$$

One degree of freedom: what should be the gradient of this linear term?

- the base functional $\rightarrow$ “KI” (Koopmans integral). Enforces IP theorem. Does not affect energy/density!

$$E_{\text{KIPZ}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \left\{ \int_0^1 \varepsilon_i(f) \, df - E_{\text{Hxc}}[n_i] \right\} \right)$$

One degree of freedom: what should be the gradient of this linear term?

- the base functional $\rightarrow$ “KI” (Koopmans integral). Enforces IP theorem. Does not affect energy/density!
- with a PZ correction $\rightarrow$ “KIPZ”

$$E_{\text{KIPZ}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \left\{ \int_0^1 \varepsilon_i(f) \, df - E_{\text{Hxc}}[n_i] \right\} \right)$$

One degree of freedom: what should be the gradient of this linear term?

- the base functional $\rightarrow$ “KI” (Koopmans integral). Enforces IP theorem. Does not affect energy/density!
- with a PZ correction $\rightarrow$ “KIPZ”

You might also see...

$$E_{\text{KIPZ}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \left\{ \int_0^1 \varepsilon_i(f) \, df - E_{\text{Hxc}}[n_i] \right\} \right)$$

One degree of freedom: what should be the gradient of this linear term?

- the base functional $\rightarrow$ “KI” (Koopmans integral). Enforces IP theorem. Does not affect energy/density!
- with a PZ correction $\rightarrow$ “KIPZ”

You might also see...

- “pKIPZ” = KIPZ Hamiltonian evaluated on the KI solution

$$E_{\text{KIPZ}}[\rho, \{f_i\}, \{\alpha_i\}] = E_{\text{DFT}}[\rho] + \sum_i \alpha_i \left(- \int_0^{f_i} \varepsilon_i(f) \, df + f_i \left\{ \int_0^1 \varepsilon_i(f) \, df - E_{\text{Hxc}}[n_i] \right\} \right)$$

One degree of freedom: what should be the gradient of this linear term?

- the base functional $\rightarrow$ “KI” (Koopmans integral). Enforces IP theorem. Does not affect energy/density!
- with a PZ correction $\rightarrow$ “KIPZ”

You might also see...

- “pKIPZ” = KIPZ Hamiltonian evaluated on the KI solution
- “K” = an earlier iteration based off half-filling rather than integer endpoints (no longer used)

Orbital-density-dependence

Orbital-density-dependence



$$-\int_0^{f_i} \varepsilon_i(f) \, df + f_i \int_0^1 \varepsilon_i(f) \, df = E_{\text{Hxc}}[\rho] + E_{\text{Hxc}}[\rho - \rho_i] + f_i(-E_{\text{Hxc}}[\rho - \rho_i] + E_{\text{Hxc}}[\rho - \rho_i + n_i])$$

Orbital-density-dependence



$$-\int_0^{f_i} \varepsilon_i(f) \, df + f_i \int_0^1 \varepsilon_i(f) \, df = E_{\text{Hxc}}[\rho] + E_{\text{Hxc}}[\rho - \rho_i] + f_i(-E_{\text{Hxc}}[\rho - \rho_i] + E_{\text{Hxc}}[\rho - \rho_i + n_i])$$

For filled orbitals with KI:

$$v_i^{\text{KI}}/\alpha_i = -E_{\text{H}}[n_i] + E_{\text{xc}}[\rho] - E_{\text{xc}}[\rho - n_i] - \int d\mathbf{r}' v_{\text{xc}}(\mathbf{r}', [\rho]) n_i(\mathbf{r}')$$

Orbital-density-dependence

Initialise with MLWFs, then (optionally) solve with CG minimisation:

G. Borghi *et al.* *Phys. Rev. B* **91**, 155112 (2015)

Orbital-density-dependence

Initialise with MLWFs, then (optionally) solve with CG minimisation:

outer loop $\varphi_i^{(n+1)} = \varphi_i^{(n)} + \Delta_i$

Orbital-density-dependence

Initialise with MLWFs, then (optionally) solve with CG minimisation:

outer loop $\varphi_i^{(n+1)} = \varphi_i^{(n)} + \Delta_i$
inner loop $\varphi_i^{(n+1)} = \sum_j U_{ij} \varphi_j^{(n)}$

Orbital-density-dependence

Initialise with MLWFs, then (optionally) solve with CG minimisation:

outer loop $\varphi_i^{(n+1)} = \varphi_i^{(n)} + \Delta_i$
inner loop $\varphi_i^{(n+1)} = \sum_j U_{ij} \varphi_j^{(n)}$

For more details see

Orbital-density-dependence



Initialise with MLWFs, then (optionally) solve with CG minimisation:

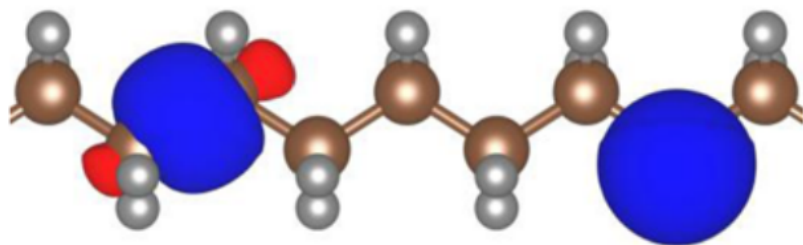
outer loop $\varphi_i^{(n+1)} = \varphi_i^{(n)} + \Delta_i$

inner loop $\varphi_i^{(n+1)} = \sum_j U_{ij} \varphi_j^{(n)}$

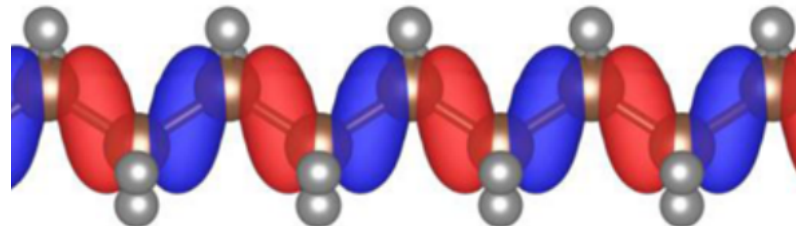
For more details see

Gives rise to a set of **minimising orbitals** (localised/variational); diagonalising at the minimum gives rise to **diagonalising orbitals** (delocalised/canonical)

Orbital-density-dependence



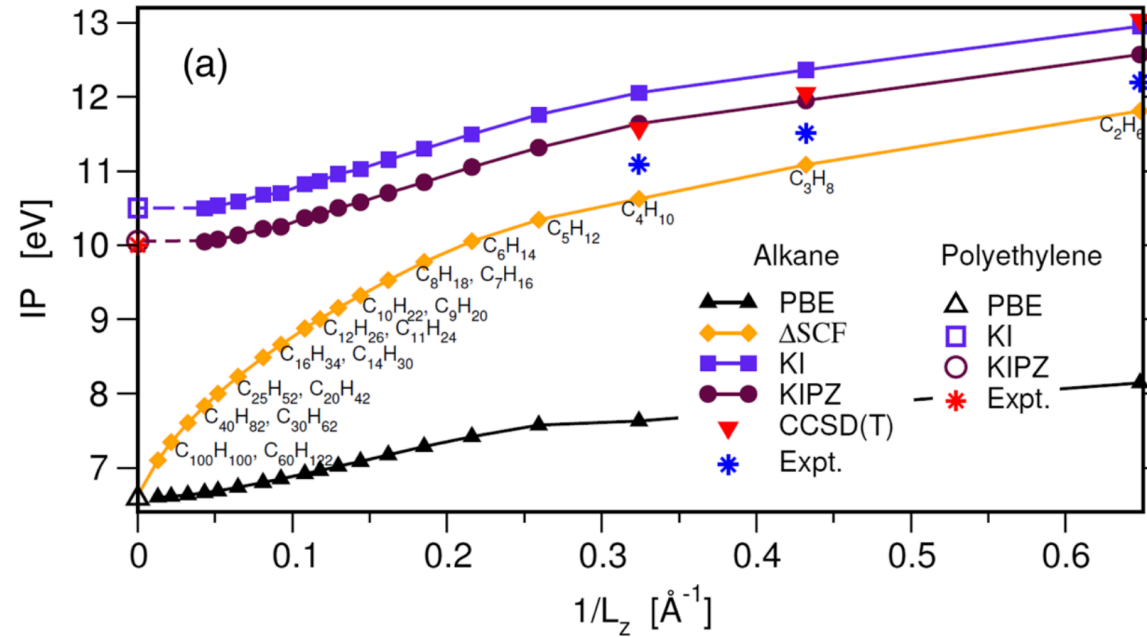
variational



canonical

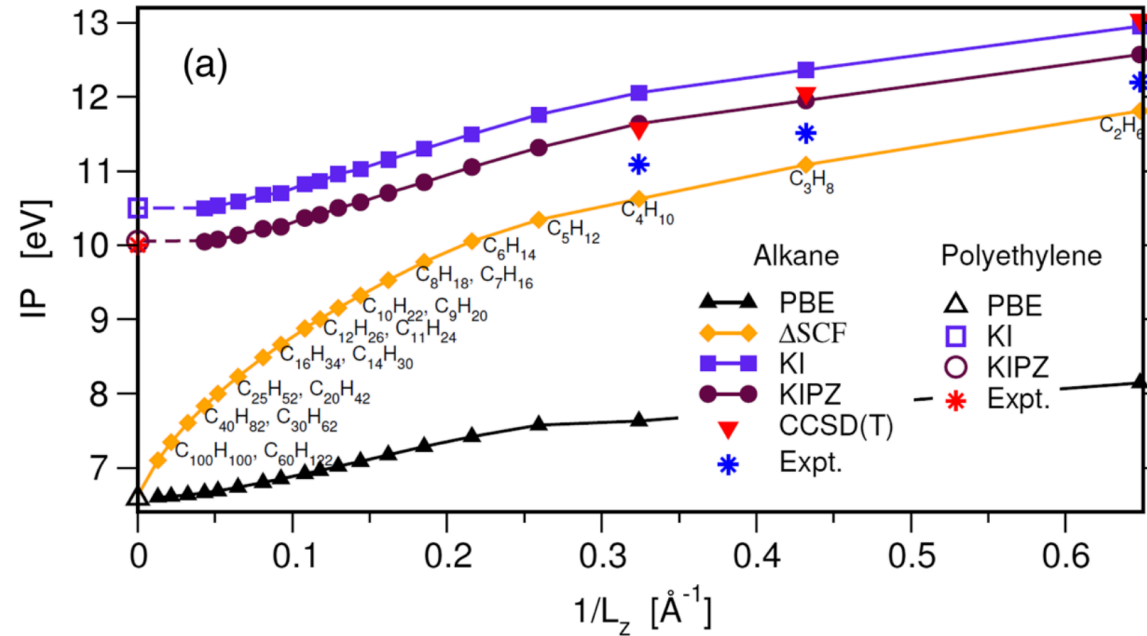
N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018)

Importance of localisation



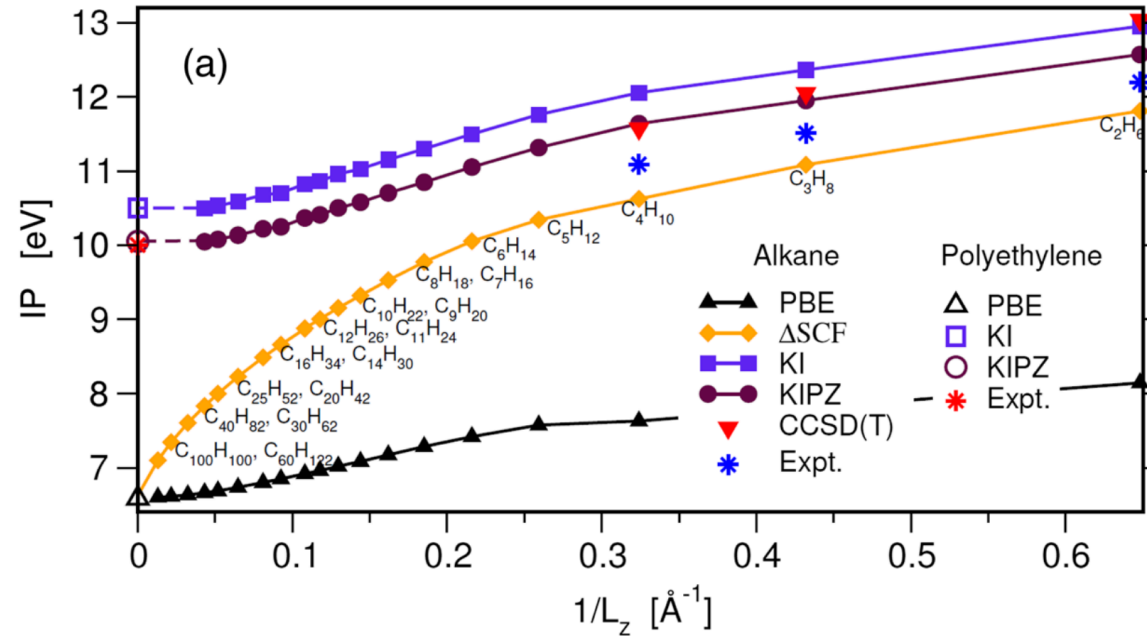
N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018)

Importance of localisation



In the bulk limit for one cell $\Delta E = E(N - \delta N) - E(N)$

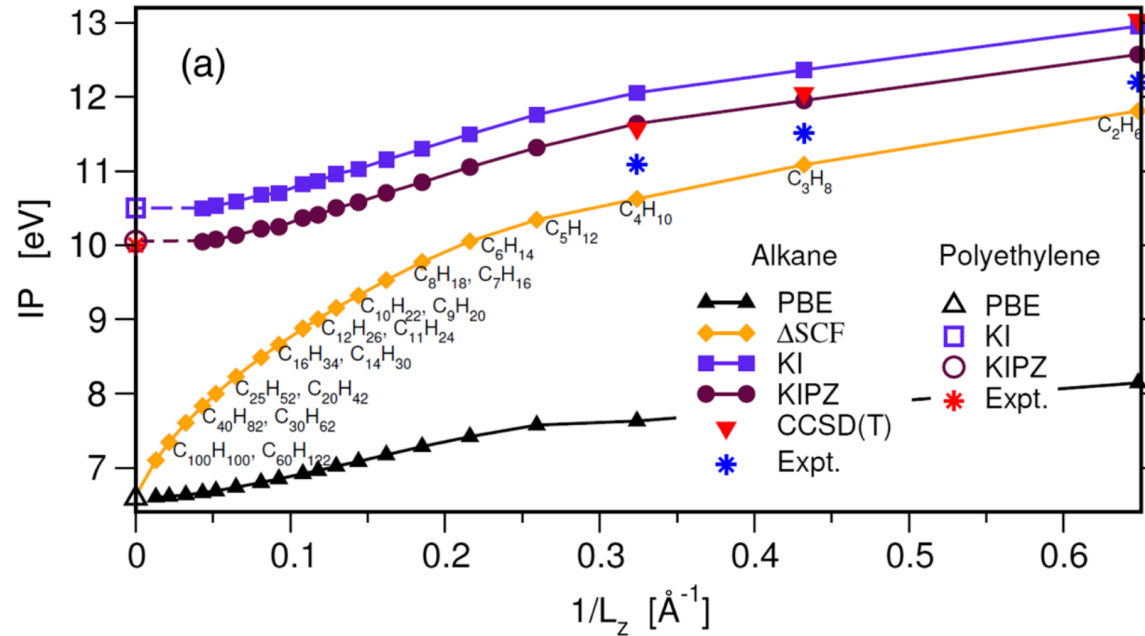
Importance of localisation



In the bulk limit for one cell $\Delta E = E(N - \delta N) - E(N)$

Across all the cells $\Delta E = \frac{1}{\delta N} (E(N - \delta N) - E(N)) = -\frac{dE}{dN} = -\varepsilon_{\text{HO}}$

Importance of localisation



$$\lim_{n_i(r) \rightarrow 0} v_i^{\text{KI}} / \alpha_i = \lim_{n_i(r) \rightarrow 0} -E_{\text{H}}[n_i] + E_{\text{xc}}[\rho] - E_{\text{xc}}[\rho - n_i] - \int d\mathbf{r}' v_{\text{xc}}(\mathbf{r}', [\rho]) n_i(\mathbf{r}') = 0$$

Orbital-density-dependence



Other features of orbital-density-dependence

A. Ferretti *et al.* *Phys. Rev. B* **89**, 195134 (2014)

Orbital-density-dependence



Other features of orbital-density-dependence

- ODD functional means that we know $\hat{H}|\varphi_i\rangle$ for variational orbitals $\{|\varphi_i\rangle\}$ but we don't know $\hat{H}$ in general

Orbital-density-dependence



Other features of orbital-density-dependence

- ODD functional means that we know $\hat{H}|\varphi_i\rangle$ for variational orbitals $\{|\varphi_i\rangle\}$ but we don't know $\hat{H}$ in general
- Practically we can often use MLWFs

Orbital-density-dependence



Other features of orbital-density-dependence

- ODD functional means that we know $\hat{H}|\varphi_i\rangle$ for variational orbitals $\{|\varphi_i\rangle\}$ but we don't know $\hat{H}$ in general
- Practically we can often use MLWFs
- a natural generalisation in the direction of spectral functional theory (as discussed already by Nicola)

Screening

- In Hartree-Fock (the original “Koopmans’ theorem”)

$$E_{\text{ee}}^{\text{HF}} = \frac{1}{2} \sum_{ij} f_i f_j \int d\mathbf{r} d\mathbf{r}' \frac{|\psi_i(\mathbf{r})|^2 |\psi_j(\mathbf{r}')|^2}{r - r'} - \frac{\psi_i^*(\mathbf{r}) \psi_j^*(\mathbf{r}') \psi_i(\mathbf{r}') \psi_j(\mathbf{r})}{r - r'}$$

- In Hartree-Fock (the original “Koopmans’ theorem”)

$$E_{\text{ee}}^{\text{HF}} = \frac{1}{2} \sum_{ij} f_i f_j \int d\mathbf{r} d\mathbf{r}' \frac{|\psi_i(\mathbf{r})|^2 |\psi_j(\mathbf{r}')|^2}{r - r'} - \frac{\psi_i^*(\mathbf{r}) \psi_j^*(\mathbf{r}') \psi_i(\mathbf{r}') \psi_j(\mathbf{r})}{r - r'}$$

- Account for screening post-hoc:

$$\frac{dE}{df_i} \approx \alpha_i \frac{\partial E}{\partial f_i}$$

- In Hartree-Fock (the original “Koopmans’ theorem”)

$$E_{\text{ee}}^{\text{HF}} = \frac{1}{2} \sum_{ij} f_i f_j \int d\mathbf{r} d\mathbf{r}' \frac{|\psi_i(\mathbf{r})|^2 |\psi_j(\mathbf{r}')|^2}{r - r'} - \frac{\psi_i^*(\mathbf{r}) \psi_j^*(\mathbf{r}') \psi_i(\mathbf{r}') \psi_j(\mathbf{r})}{r - r'}$$

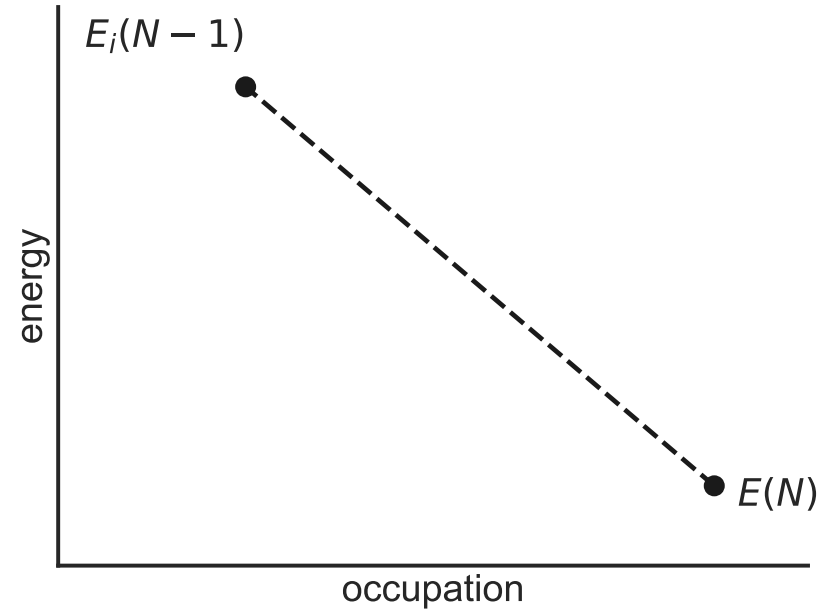
- Account for screening post-hoc:

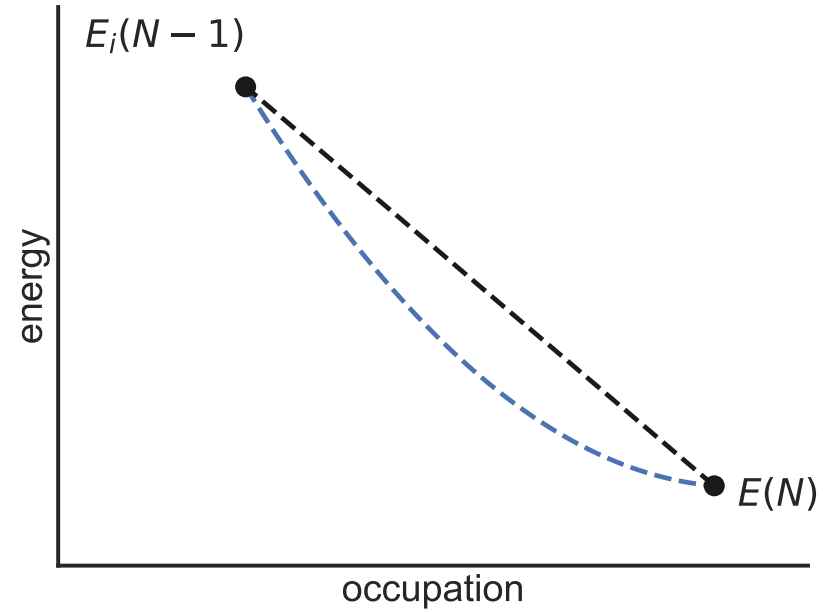
$$\frac{dE}{df_i} \approx \alpha_i \frac{\partial E}{\partial f_i}$$

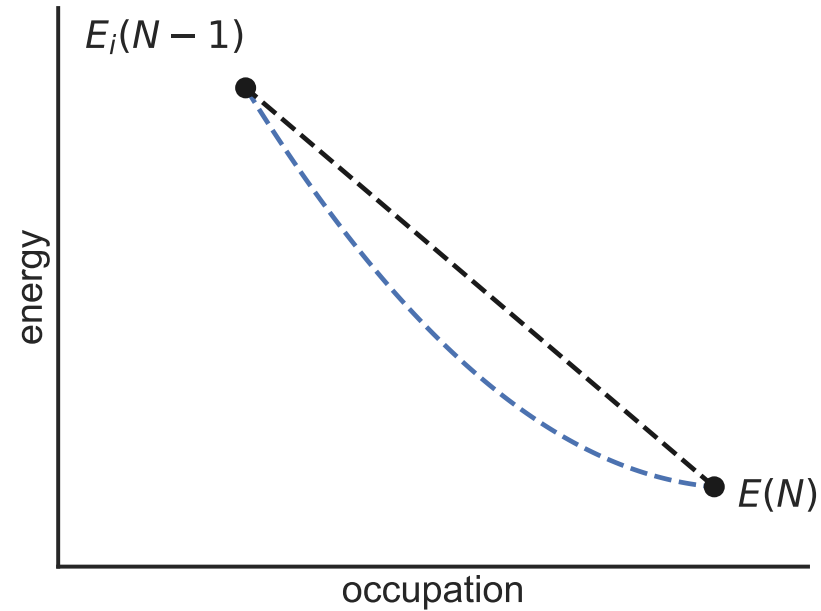
- How to choose an appropriate value for α_i ? Return to the original idea of Koopmans functionals:

$$\varepsilon_i^{\text{Koopmans}} = E_i(N - 1) - E(N)$$

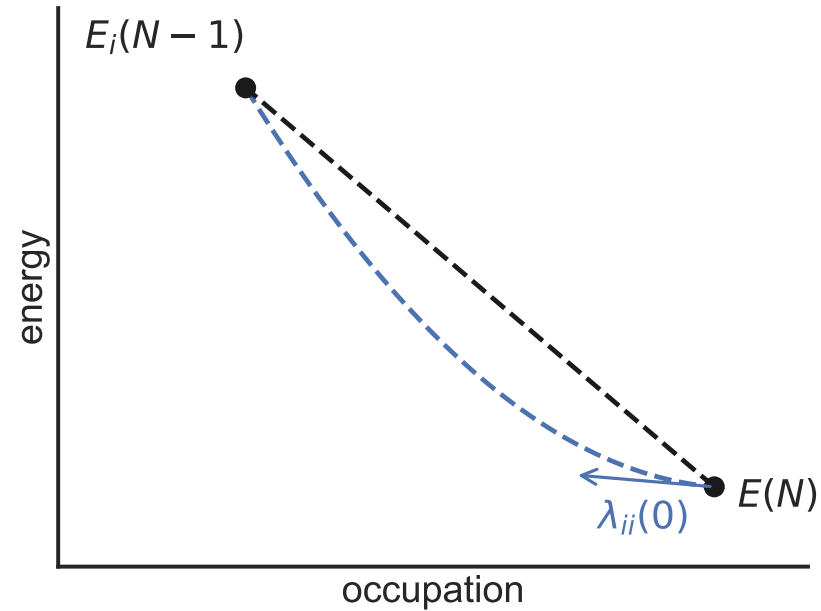




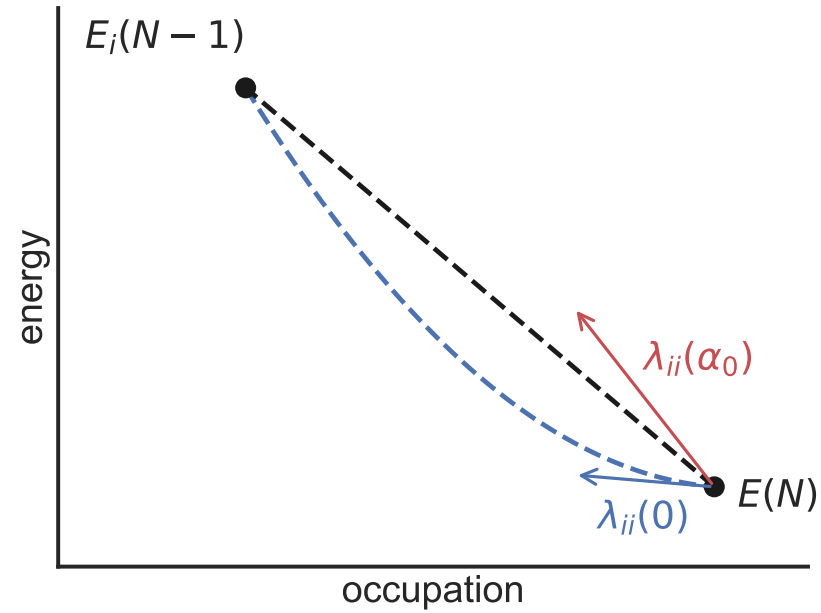




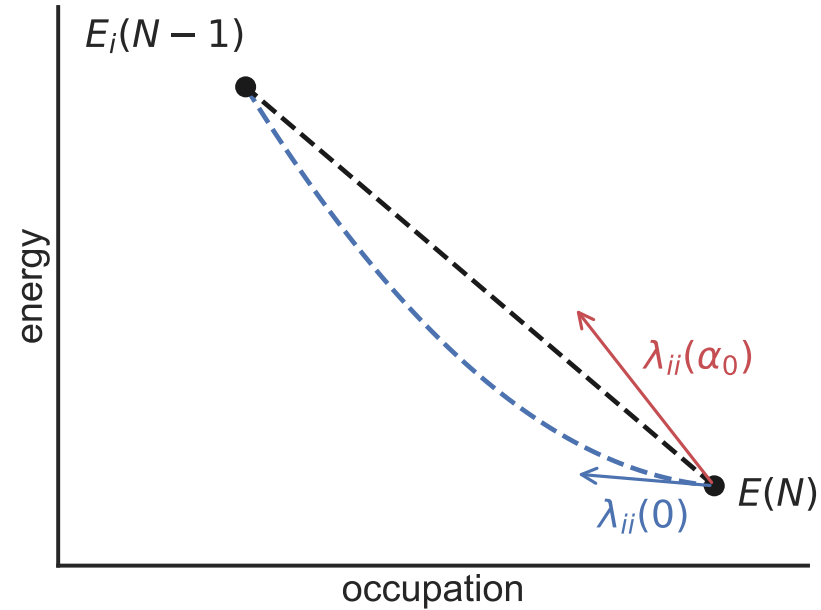
$$\lambda_{ii}(\alpha) \equiv \langle \varphi_i | \hat{h}^{\text{DFT}} + \alpha \hat{v}^{\text{Koopmans}} | \varphi_i \rangle = \left. \frac{dE^{\text{Koopmans}}}{df_i} \right|_{f_i=s}$$



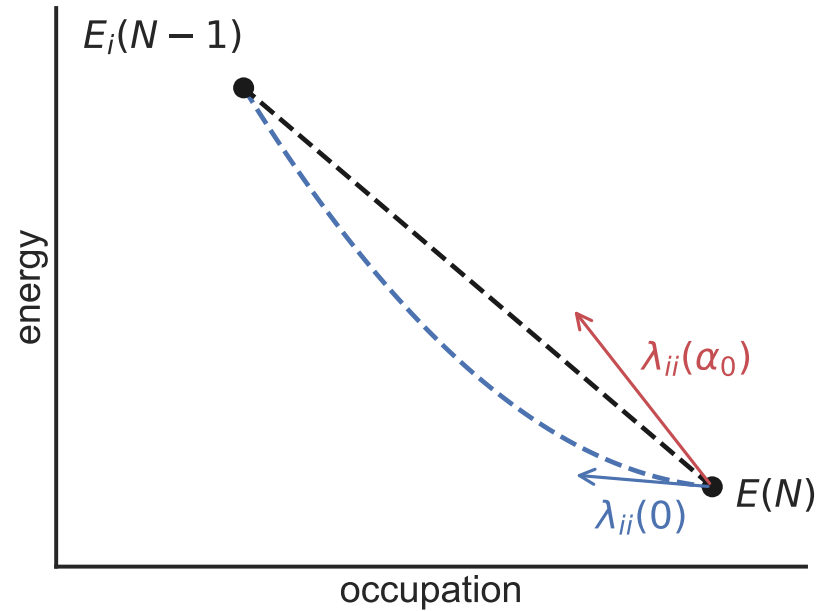
$$\lambda_{ii}(\alpha) \equiv \left\langle \varphi_i \left| \hat{h}^{\text{DFT}} + \alpha \hat{v}^{\text{Koopmans}} \right| \varphi_i \right\rangle = \left. \frac{dE^{\text{Koopmans}}}{df_i} \right|_{f_i=s}$$



$$\lambda_{ii}(\alpha) \equiv \left\langle \varphi_i \left| \hat{h}^{\text{DFT}} + \alpha \hat{v}^{\text{Koopmans}} \right| \varphi_i \right\rangle = \left. \frac{dE^{\text{Koopmans}}}{df_i} \right|_{f_i=s}$$



Given this, how to work out the ideal α ?



Given this, how to work out the ideal α ?

$$\alpha_{n+1} = \alpha_n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha_n) - \lambda_{ii}(0)}$$

$$\alpha^{n+1} = \alpha^n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha^n) - \lambda_{ii}(0)}$$

↑
screening parameter

total energy with electron
removed from orbital i

↓

$$\alpha^{n+1} = \alpha^n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha^n) - \lambda_{ii}(0)}$$

↑

screening parameter

$$\alpha^{n+1} = \alpha^n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha^n) - \lambda_{ii}(0)}$$

Diagram illustrating the iterative calculation of the screening parameter α for orbital i .

The equation is:

$$\alpha^{n+1} = \alpha^n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha^n) - \lambda_{ii}(0)}$$

Annotations:

- $E_i(N-1)$: total energy with electron removed from orbital i
- $E(N)$: total energy of neutral system
- $\lambda_{ii}(0)$: screening parameter at $\alpha=0$
- $\lambda_{ii}(\alpha^n)$: screening parameter at α^n
- α^n : screening parameter (indicated by an arrow from the text "screening parameter" below)

$$\alpha^{n+1} = \alpha^n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha^n) - \lambda_{ii}(0)}$$

Diagram illustrating the screening parameter α^{n+1} calculation:

- $E_i(N-1)$: total energy with electron removed from orbital i
- $E(N)$: total energy of neutral system
- $\lambda_{ii}(0)$: expectation value of $\hat{H}^{\text{Koopmans}}$
- $\lambda_{ii}(\alpha^n)$: expectation value of $\hat{H}^{\text{Koopmans}}$ at screening parameter α^n
- α^n : screening parameter

$$\alpha^{n+1} = \alpha^n \frac{E_i(N-1) - E(N) - \lambda_{ii}(0)}{\lambda_{ii}(\alpha^n) - \lambda_{ii}(0)}$$

Diagram illustrating the screening parameter α^{n+1} calculation:

- $E_i(N-1)$: total energy with electron removed from orbital i
- $E(N)$: total energy of neutral system
- $\lambda_{ii}(0)$: expectation value of $\hat{H}^{\text{Koopmans}}$
- $\lambda_{ii}(\alpha^n)$: expectation value of $\hat{H}^{\text{DFT}}$
- α^n : screening parameter

Screening via DFPT

How can we avoid explicit charged defect calculations in a supercell?

N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022)

R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

Screening via DFPT

How can we avoid explicit charged defect calculations in a supercell?

Reformulate in terms of DFPT...

$$\alpha_i = 1 + \frac{\langle v_{\text{pert}}^i | \Delta^i n \rangle}{\langle n_i | v_{\text{pert}}^i \rangle}$$

N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022)

R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

Screening via DFPT

How can we avoid explicit charged defect calculations in a supercell?

Reformulate in terms of DFPT...

$$\alpha_i = 1 + \frac{\langle v_{\text{pert}}^i \mid \Delta^i n \rangle}{\langle n_i \mid v_{\text{pert}}^i \rangle}$$

... in reciprocal space

$$\alpha_{0i} = 1 + \frac{\sum_{\mathbf{q}} \langle v_{\text{pert},\mathbf{q}}^{0i} \mid \Delta_{\mathbf{q}}^{0i} n \rangle}{\sum_{\mathbf{q}} \langle n_{\mathbf{q}}^{0i} \mid v_{\text{pert},\mathbf{q}}^{0i} \rangle}$$

N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022)

R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

Screening via DFPT

How can we avoid explicit charged defect calculations in a supercell?

Reformulate in terms of DFPT...

$$\alpha_i = 1 + \frac{\langle v_{\text{pert}}^i \mid \Delta^i n \rangle}{\langle n_i \mid v_{\text{pert}}^i \rangle}$$

... in reciprocal space

$$\alpha_{0i} = 1 + \frac{\sum_{\mathbf{q}} \langle v_{\text{pert},\mathbf{q}}^{0i} \mid \Delta_{\mathbf{q}}^{0i} n \rangle}{\sum_{\mathbf{q}} \langle n_{\mathbf{q}}^{0i} \mid v_{\text{pert},\mathbf{q}}^{0i} \rangle}$$

N.B. even for the supercell, we can still reconstruct a band structure

N. Colonna *et al.* *J. Chem. Theory Comput.* **15**, 1905–1914 (2019)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022)

R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

To summarise...

When performing a Koopmans calculation, you must decide...

To summarise...

When performing a Koopmans calculation, you must decide...

- which flavour (KI, KIPZ)?

To summarise...

When performing a Koopmans calculation, you must decide...

- which flavour (KI, KIPZ)?
- how are we treating the orbital-density-dependence?
 - how are we initialising our variational orbitals? (N.B. depends on the flavour!)
 - are we going to explicitly minimise the ODD?

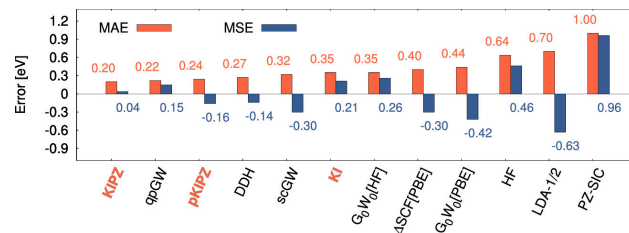
To summarise...

When performing a Koopmans calculation, you must decide...

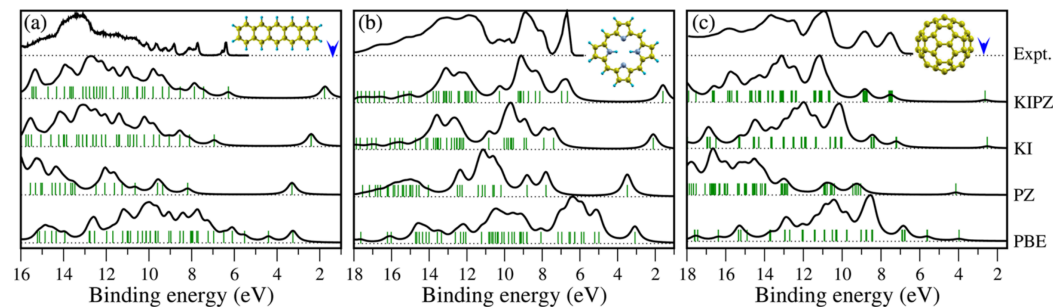
- which flavour (KI, KIPZ)?
- how are we treating the orbital-density-dependence?
 - how are we initialising our variational orbitals? (N.B. depends on the flavour!)
 - are we going to explicitly minimise the ODD?
- how are we calculating the screening parameters? (finite differences, DFPT, ML...)

Results

Ionisation potentials of 100 molecules cf. CCSD(T)

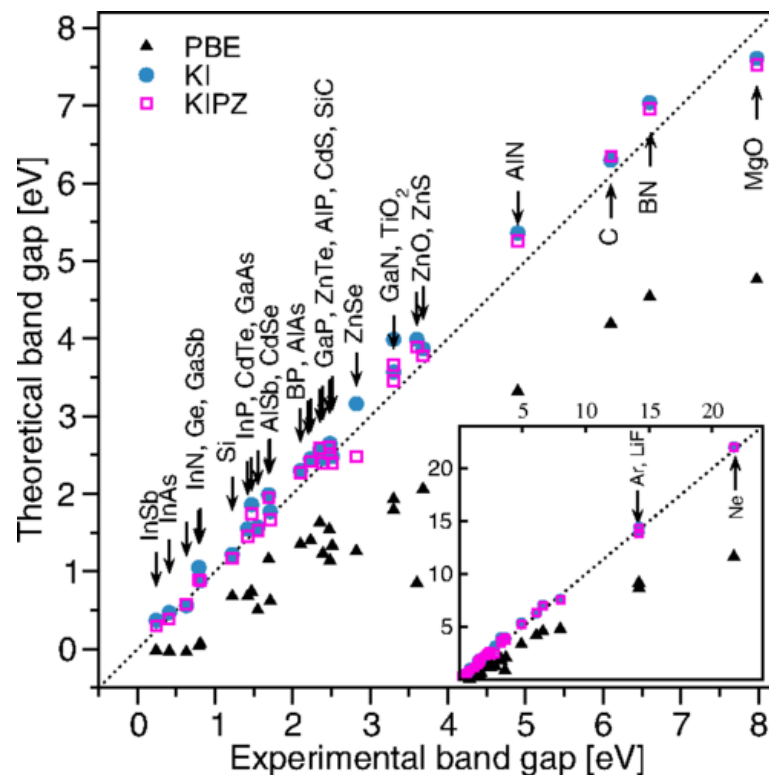


Ultraviolet photoemission spectra



N. Colonna *et al.* *J. Chem. Theory Comput.* **14**, 2549–2557 (2018), N. L. Nguyen *et al.* *Phys. Rev. Lett.* **114**, 166405 (2015)

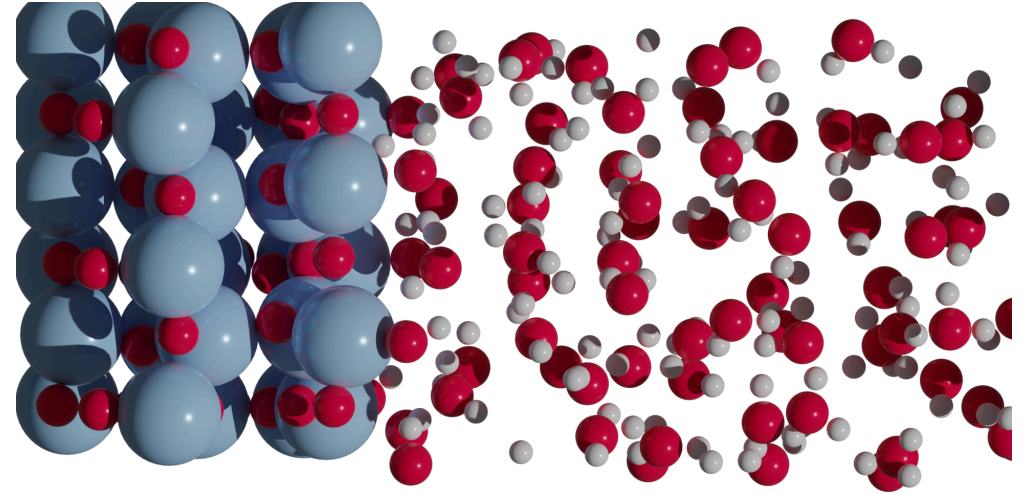
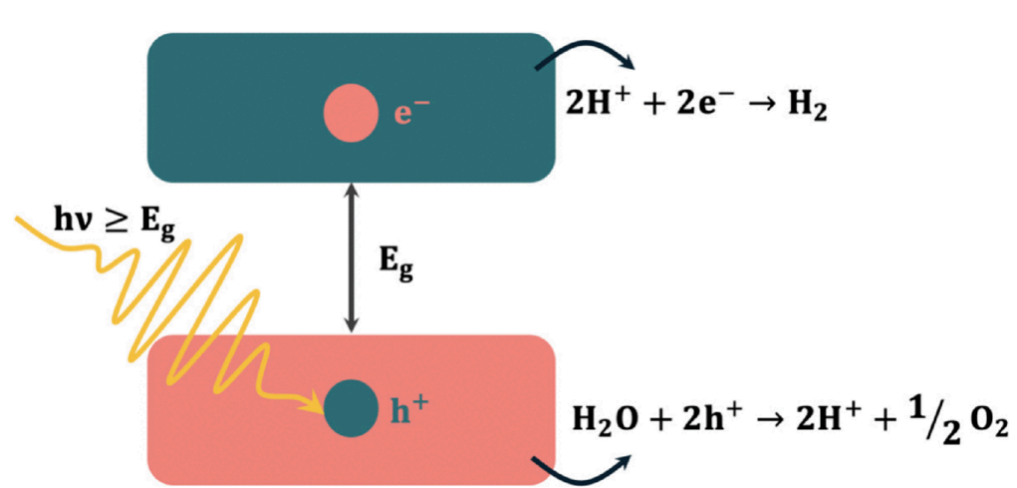
Koopmans functionals: results for solids



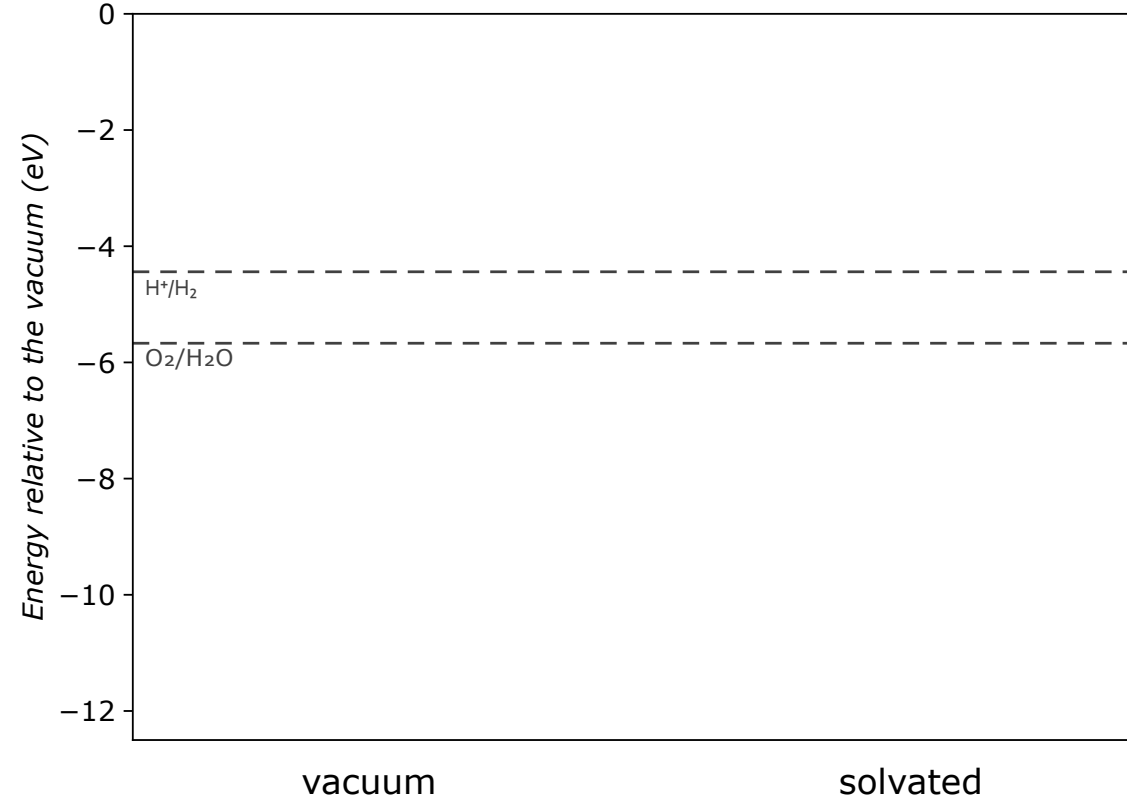
Mean absolute error (eV) across prototypical semiconductors and insulators

	PBE	G_0W_0	KI	KIPZ	QSG $\tilde{W}$
E_{gap}	2.54	0.56	0.27	0.22	0.18
IP	1.09	0.39	0.19	0.21	0.49

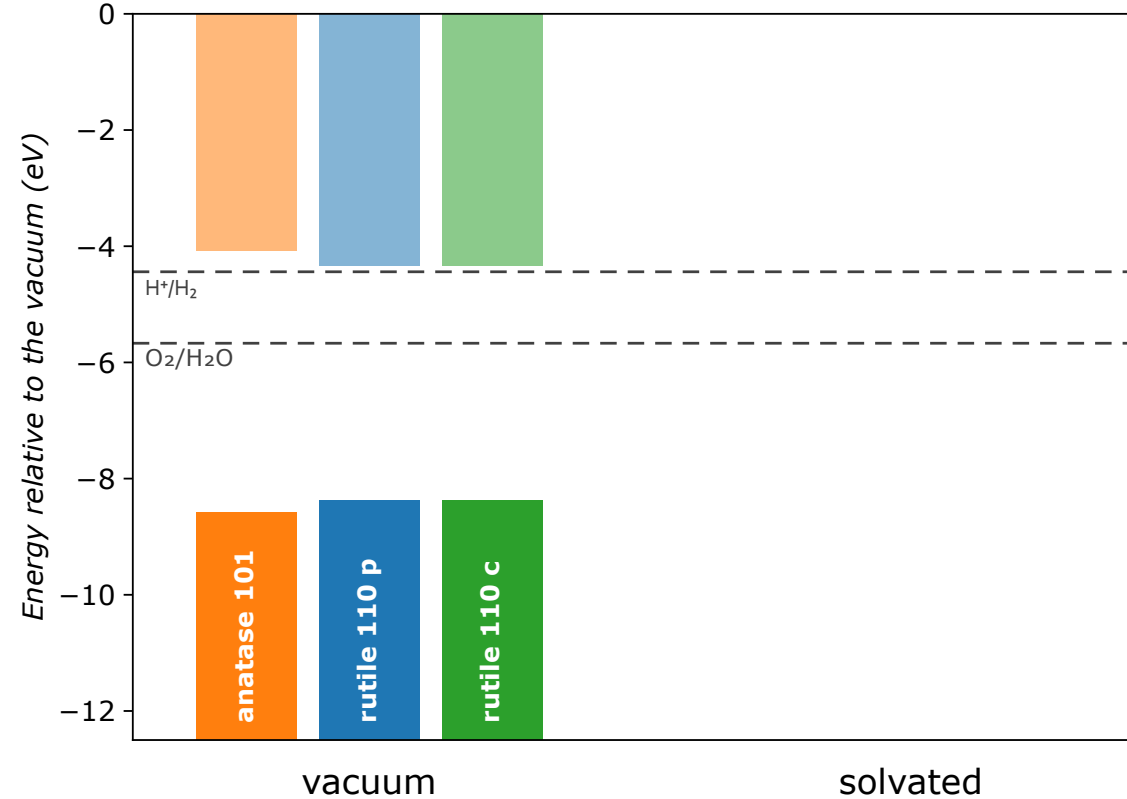
Photocatalytic water-splitting



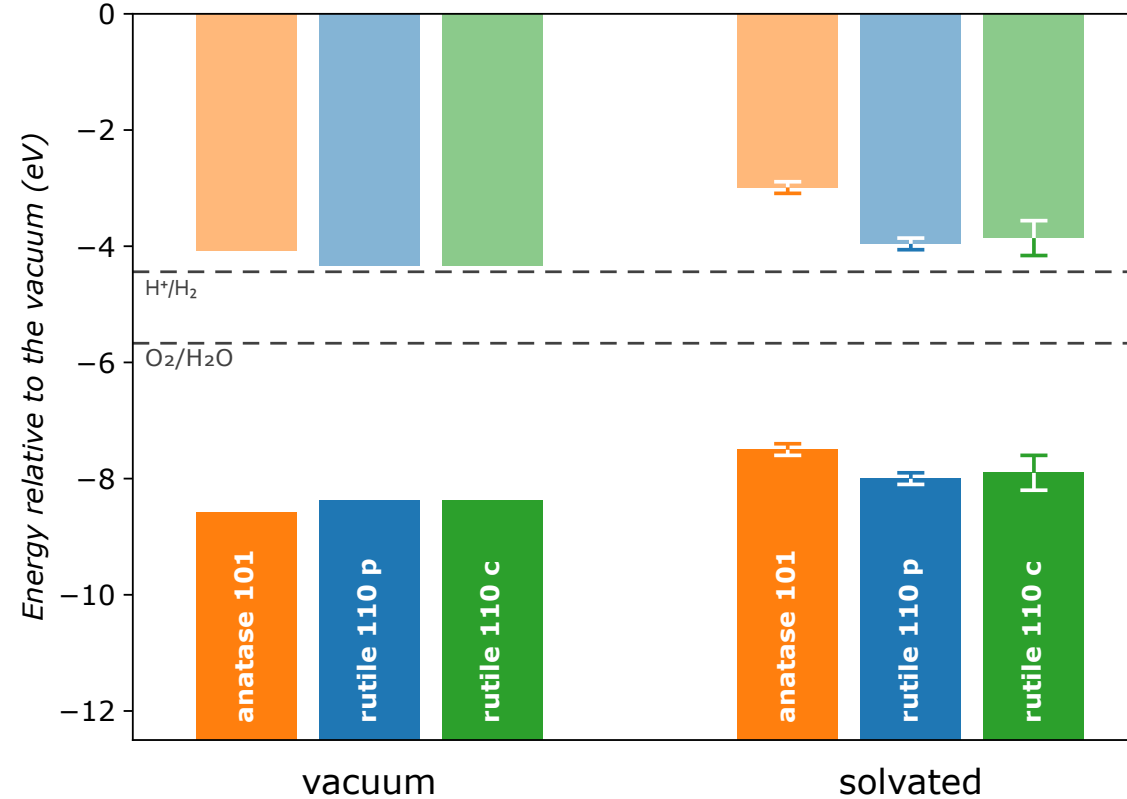
Band alignment for water-splitting



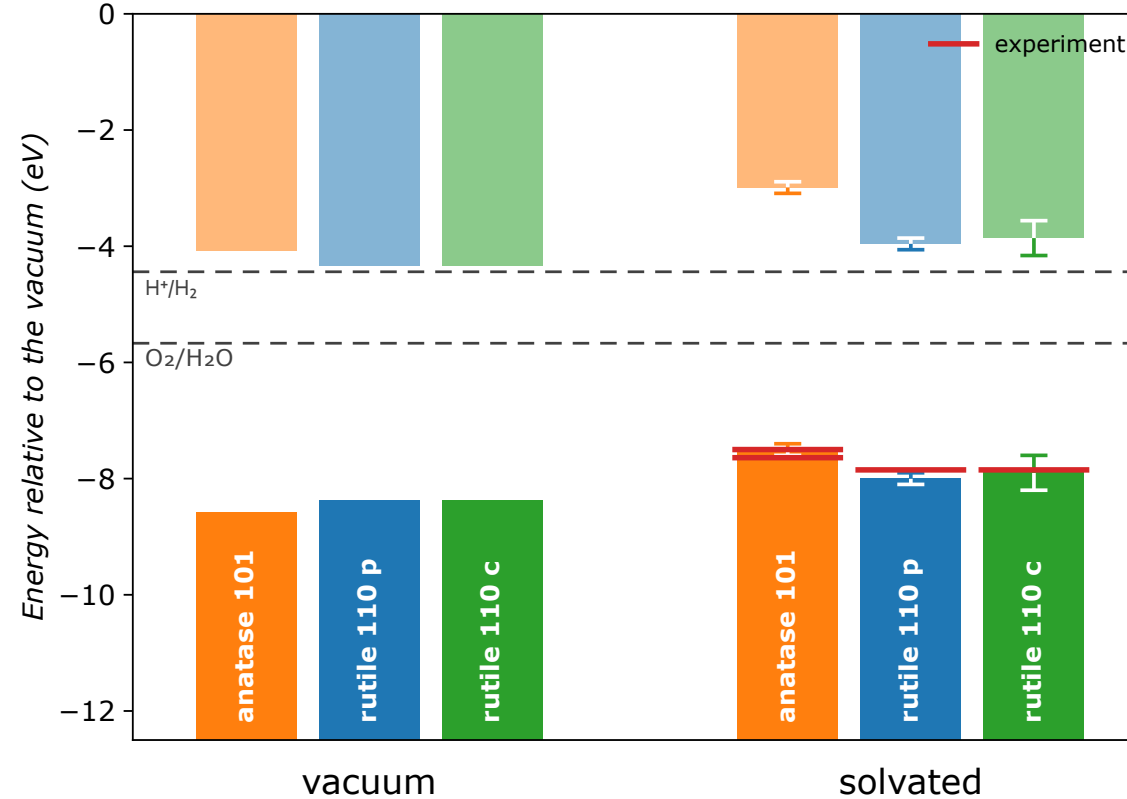
Band alignment for water-splitting



Band alignment for water-splitting

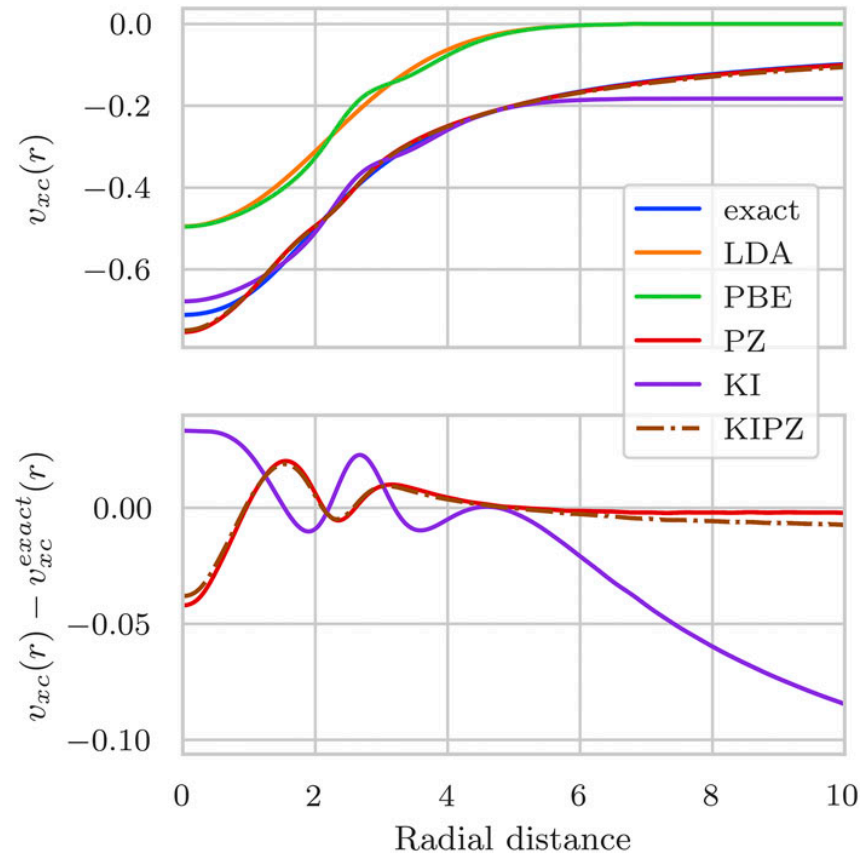


Band alignment for water-splitting



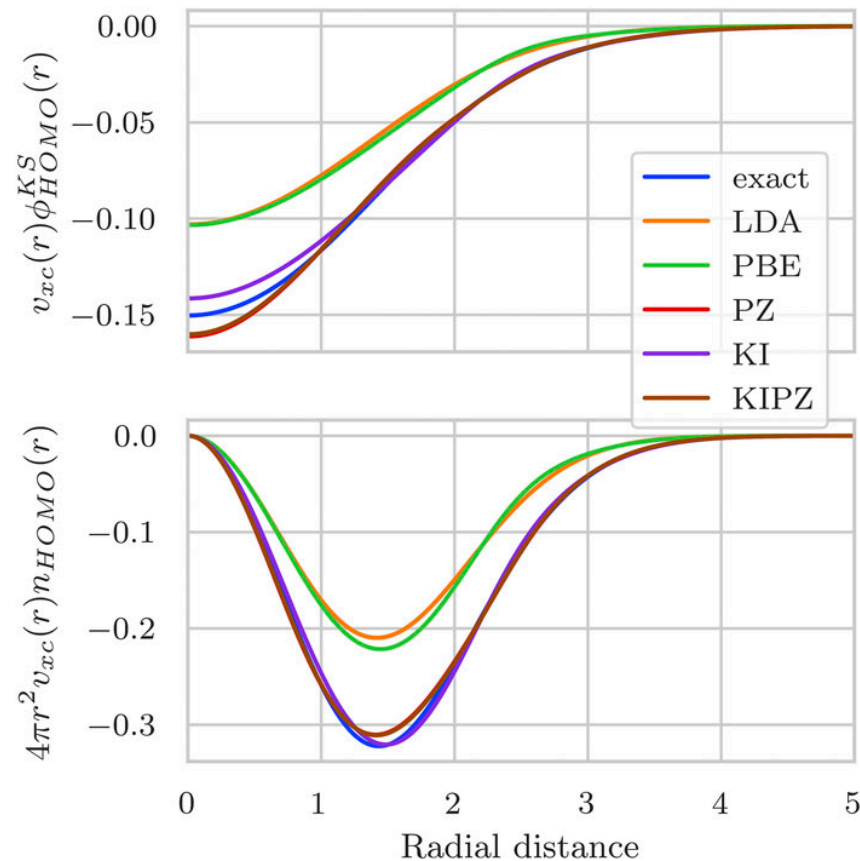
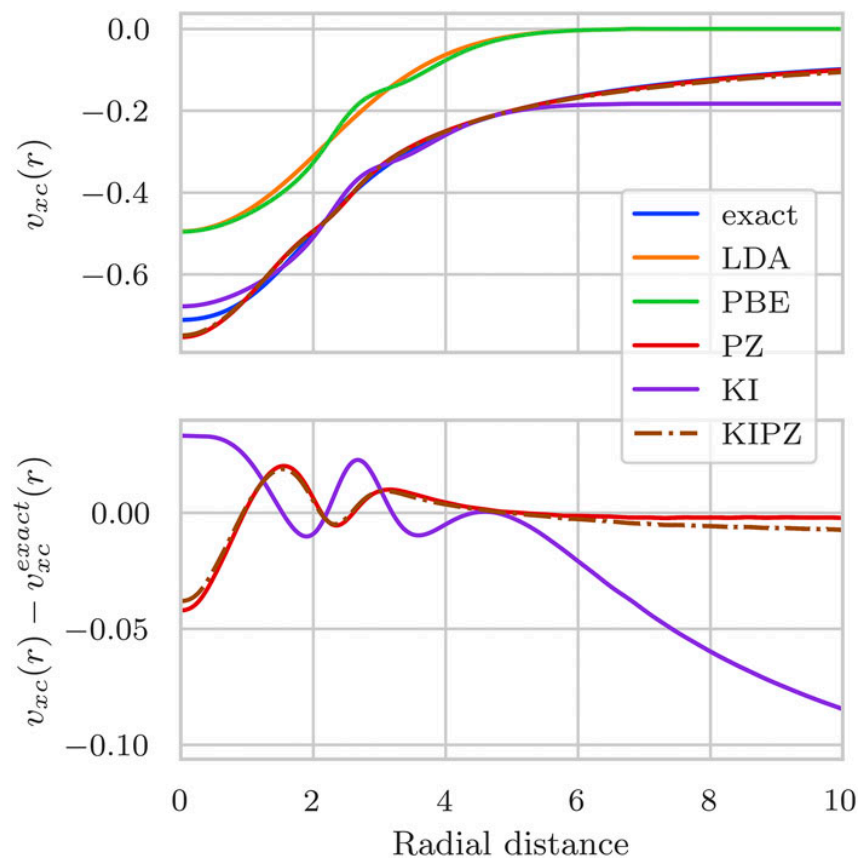
Toy systems

For Hooke's atom (two electrons in a harmonic confining potential with Coulombic repulsion)



Toy systems

For Hooke's atom (two electrons in a harmonic confining potential with Coulombic repulsion)



Koopmans functionals: caveats



Koopmans functionals: caveats

- restricted to systems with a non-zero band gap

Koopmans functionals: caveats



- restricted to systems with a non-zero band gap
- empty state localization in the bulk limit

Koopmans functionals: caveats



- restricted to systems with a non-zero band gap
- empty state localization in the bulk limit
- can potentially break the crystal point group symmetry

Resonance with other efforts

- Wannier transition-state method of Anisimov and Kozhevnikov
- Optimally tuned hybrid functionals of Kronik, Pasquarello, and others (refer back to Leeor's talk on Wednesday)
- Ensemble DFT of Kronik and co-workers
- Koopmans-Wannier of Wang and co-workers
- Dielectric-dependent hybrid functionals of Galli and co-workers
- LOSC functionals of Yang and co-workers

V. I. Anisimov *et al. Phys. Rev. B* **72**, 75125 (2005)

L. Kronik *et al. J. Chem. Theory Comput.* **8**, 1515–1531 (2012), D. Wing *et al. Proc. Natl. Acad. Sci.* **118**, e2104556118 (2021)

E. Kraisler *et al. Phys. Rev. Lett.* **110**, 126403 (2013)

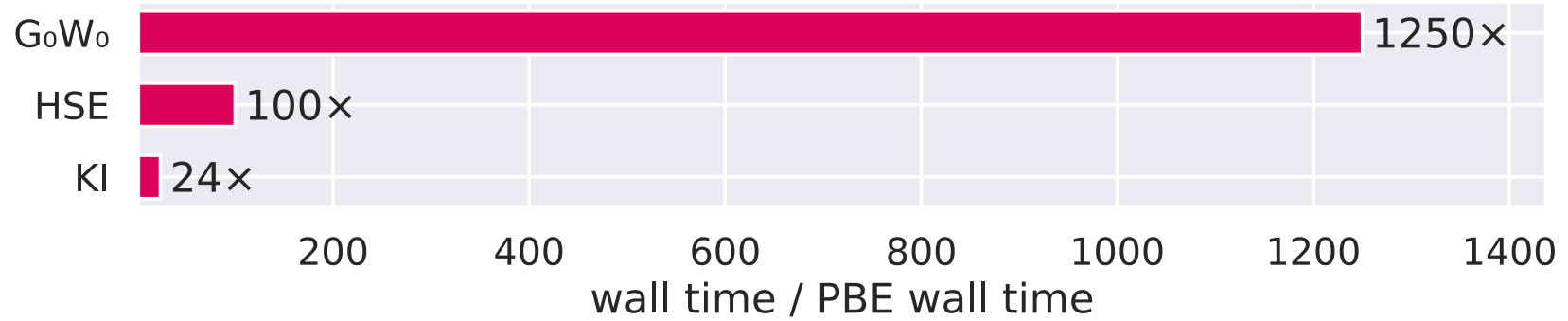
J. Ma *et al. Sci. Rep.* **6**, 24924 (2016)

J. H. Skone *et al. Phys. Rev. B* **93**, 235106 (2016)

C. Li *et al. Natl. Sci. Rev.* **5**, 203–215 (2018)

Computational cost and scaling

Computational cost and scaling



Computational cost and scaling

The vast majority of the computational cost: determining screening parameters

$$\alpha_i = \frac{\langle n_i | \varepsilon^{-1} f_{\text{Hxc}} | n_i \rangle}{\langle n_i | f_{\text{Hxc}} | n_i \rangle}$$

N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018), R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022), N. Colonna *et al.* *J. Chem. Theory Comput.* **14**, 2549–2557 (2018)

Computational cost and scaling

The vast majority of the computational cost: determining screening parameters

$$\alpha_i = \frac{\langle n_i | \varepsilon^{-1} f_{\text{Hxc}} | n_i \rangle}{\langle n_i | f_{\text{Hxc}} | n_i \rangle}$$

- a local measure of screening of electronic interactions

N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018), R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022), N. Colonna *et al.* *J. Chem. Theory Comput.* **14**, 2549–2557 (2018)

Computational cost and scaling

The vast majority of the computational cost: determining screening parameters

$$\alpha_i = \frac{\langle n_i | \varepsilon^{-1} f_{\text{Hxc}} | n_i \rangle}{\langle n_i | f_{\text{Hxc}} | n_i \rangle}$$

- a local measure of screening of electronic interactions
- one screening parameter per orbital
- must be computed *ab initio* via...

N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018), R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022), N. Colonna *et al.* *J. Chem. Theory Comput.* **14**, 2549–2557 (2018)

Computational cost and scaling

The vast majority of the computational cost: determining screening parameters

$$\alpha_i = \frac{\langle n_i | \varepsilon^{-1} f_{\text{Hxc}} | n_i \rangle}{\langle n_i | f_{\text{Hxc}} | n_i \rangle}$$

- a local measure of screening of electronic interactions
- one screening parameter per orbital
- must be computed *ab initio* via...
 - ΔSCF : embarrassingly parallel steps which each cost $\mathcal{O}(N_{\text{SC}}^3) \sim \mathcal{O}(N_{\mathbf{k}}^3 N^3)$

N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018), R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022), N. Colonna *et al.* *J. Chem. Theory Comput.* **14**, 2549–2557 (2018)

Computational cost and scaling

The vast majority of the computational cost: determining screening parameters

$$\alpha_i = \frac{\langle n_i | \varepsilon^{-1} f_{\text{Hxc}} | n_i \rangle}{\langle n_i | f_{\text{Hxc}} | n_i \rangle}$$

- a local measure of screening of electronic interactions
- one screening parameter per orbital
- must be computed *ab initio* via...
 - ΔSCF : embarrassingly parallel steps which each cost $\mathcal{O}(N_{\text{SC}}^3) \sim \mathcal{O}(N_{\mathbf{k}}^3 N^3)$
 - DFPT: $\mathcal{O}(N_{\mathbf{k}}^2 N^3)$

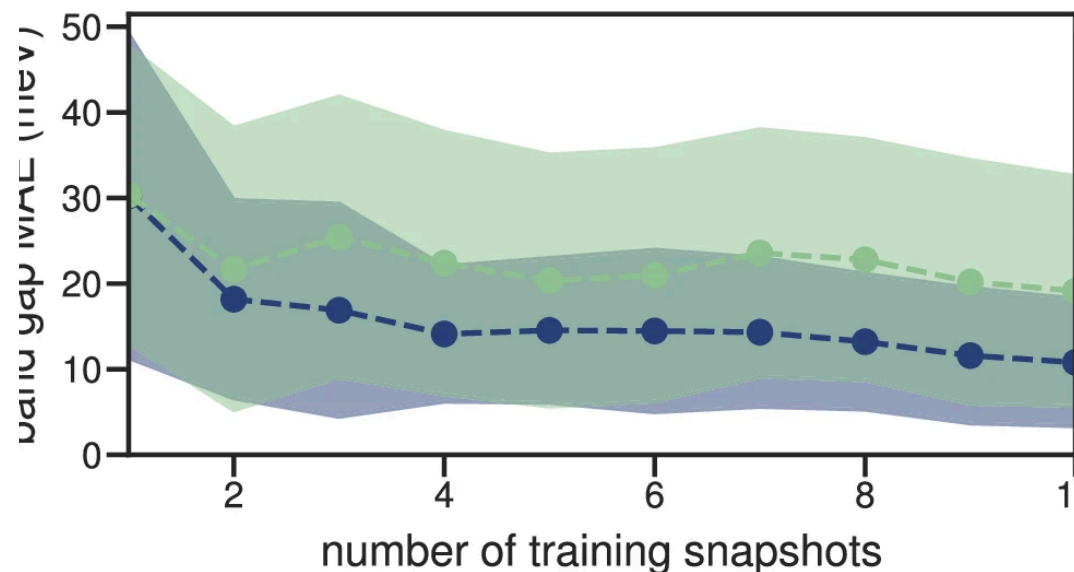
N. L. Nguyen *et al.* *Phys. Rev. X* **8**, 21051 (2018), R. De Gennaro *et al.* *Phys. Rev. B* **106**, 35106 (2022)

N. Colonna *et al.* *J. Chem. Theory Comput.* **18**, 5435–5448 (2022), N. Colonna *et al.* *J. Chem. Theory Comput.* **14**, 2549–2557 (2018)

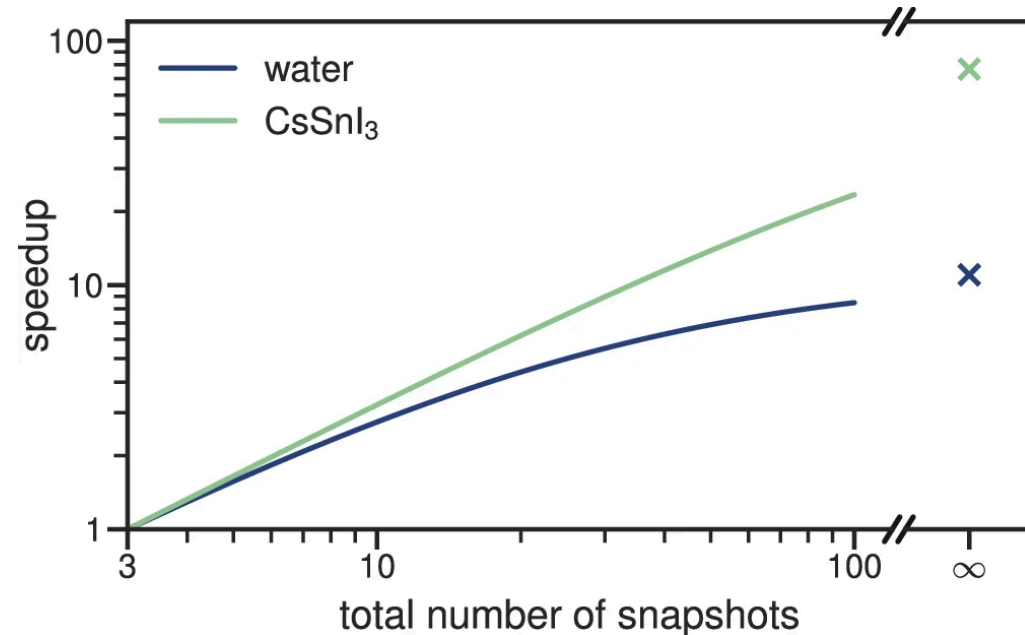
Machine-learned electronic screening



—●— water —●— CsSnI₃



accurate to within $\mathcal{O}(10 \text{ meV})$ *cf.* typical band gap accuracy of $\mathcal{O}(100 \text{ meV})$

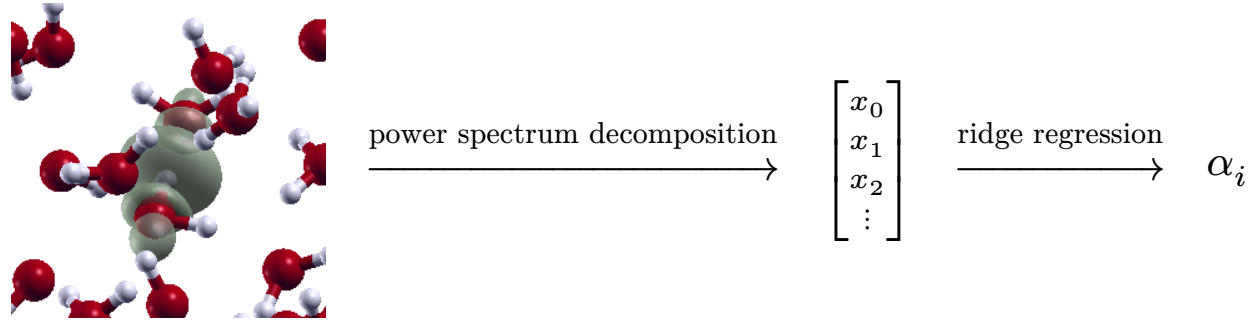


speedup of $\mathcal{O}(10)$ to $\mathcal{O}(100)$

Machine-learned electronic screening



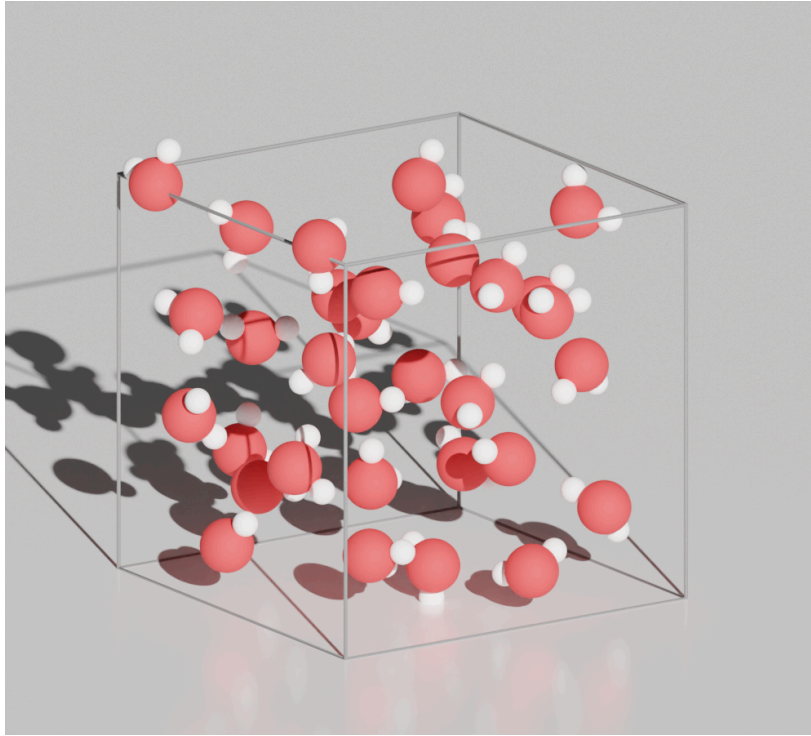
Machine-learned electronic screening



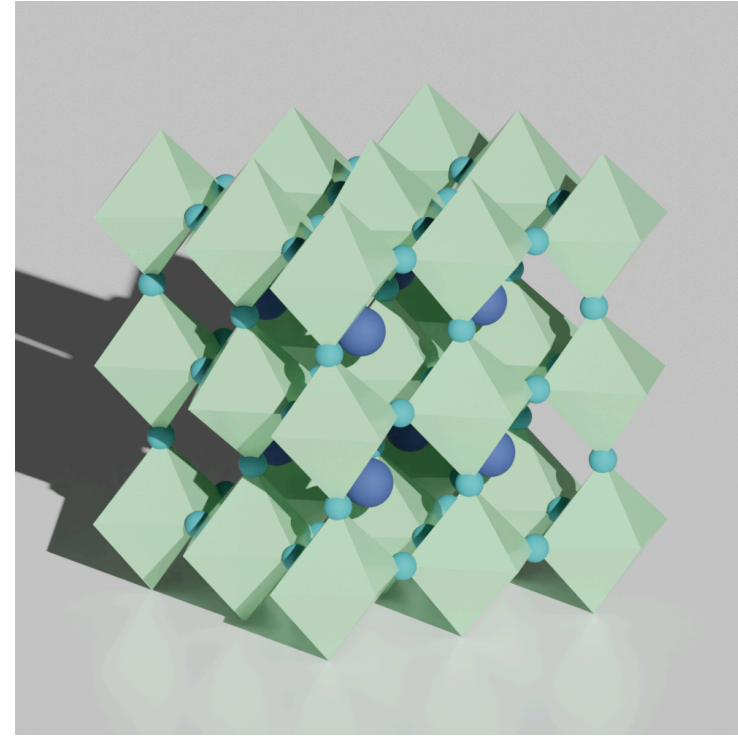
$$c_{nlm,k}^i = \int d\mathbf{r} g_{nl}(r) Y_{lm}(\theta, \varphi) n^i(\mathbf{r} - \mathbf{R}^i)$$

$$p_{n_1 n_2 l, k_1 k_2}^i = \pi \sqrt{\frac{8}{2l+1}} \sum_m c_{n_1 l m, k_1}^{i*} c_{n_2 l m, k_2}^i$$

Machine-learned electronic screening



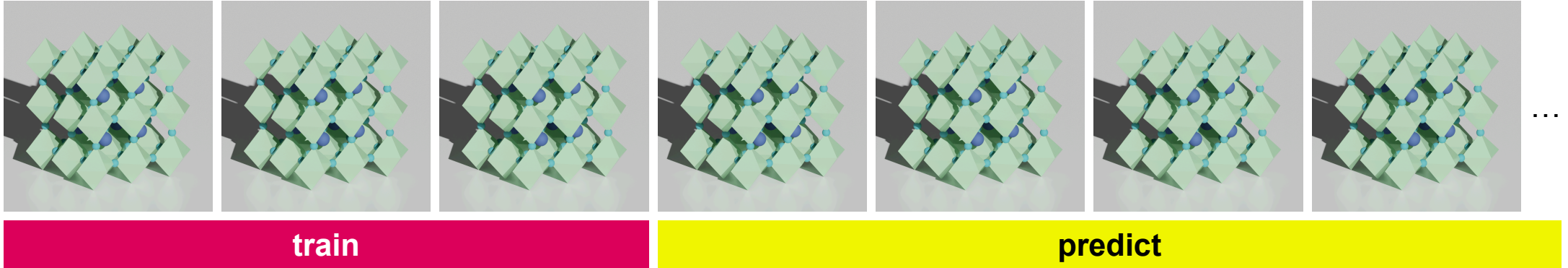
water



CsSnI_3

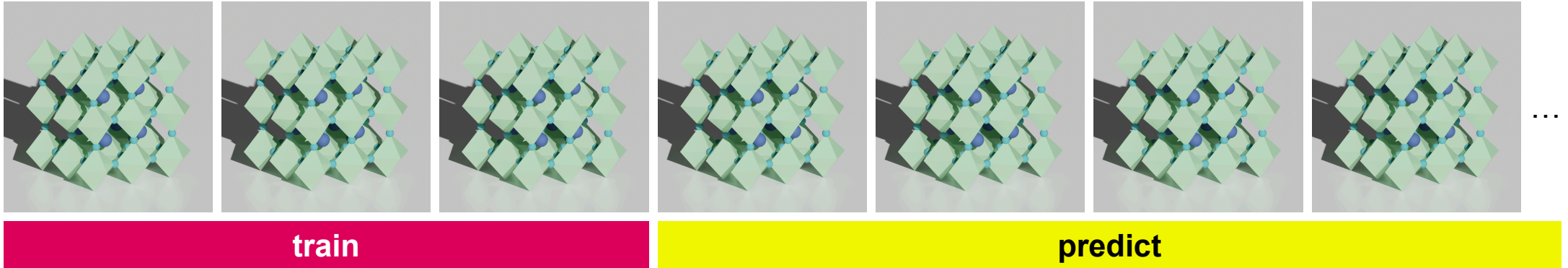
Machine-learned electronic screening

The use-case



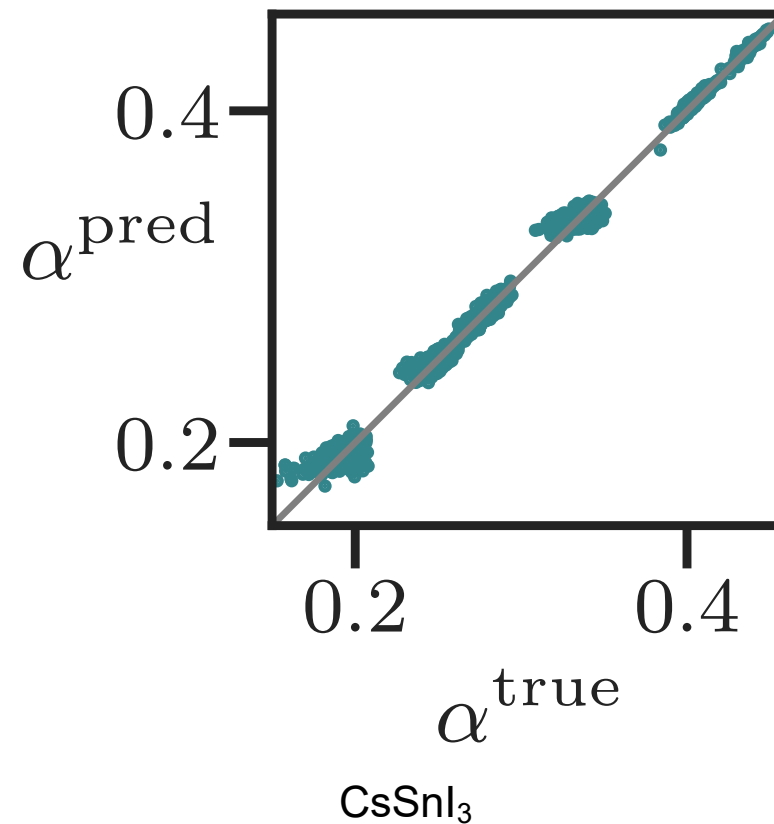
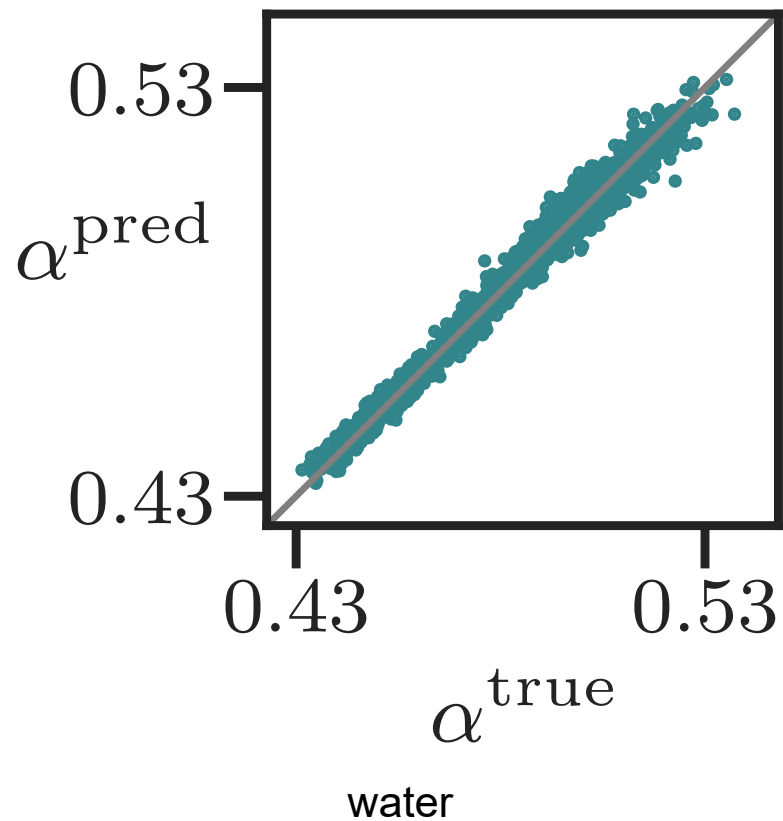
Machine-learned electronic screening

The use-case

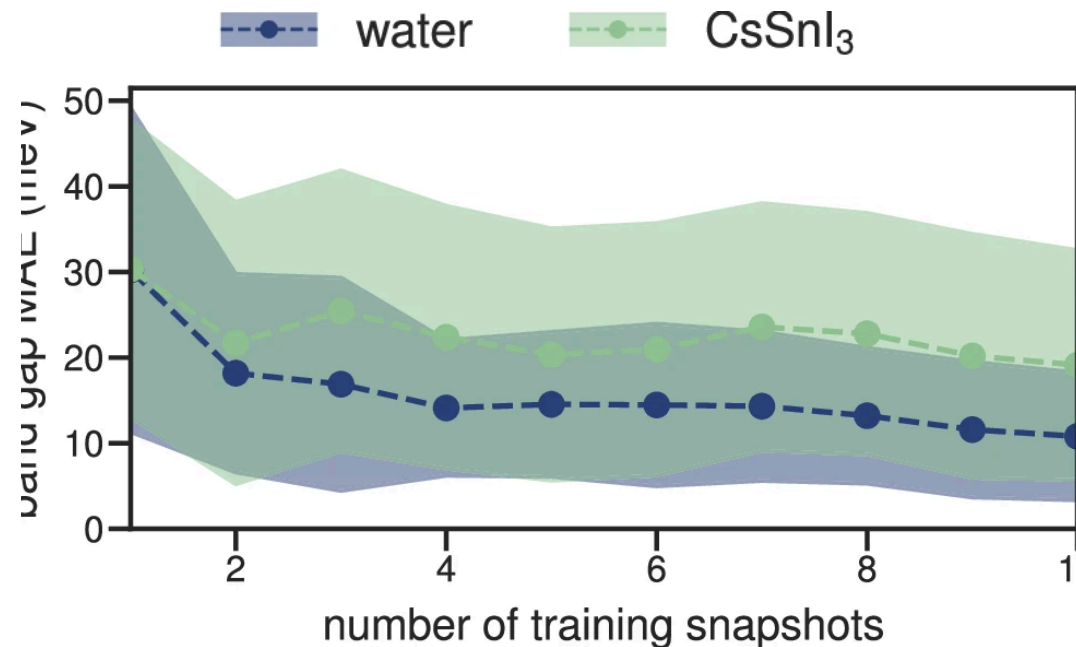


N.B. not a general model

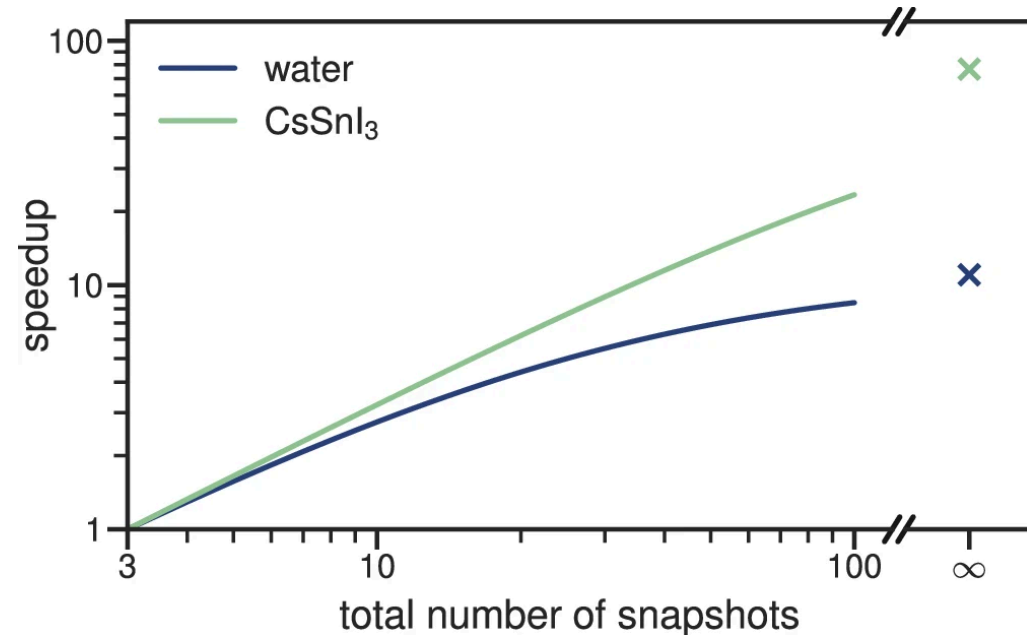
Machine-learned electronic screening



Machine-learned electronic screening



accurate to within $\mathcal{O}(10 \text{ meV})$ *cf.* typical band gap accuracy of $\mathcal{O}(100 \text{ meV})$



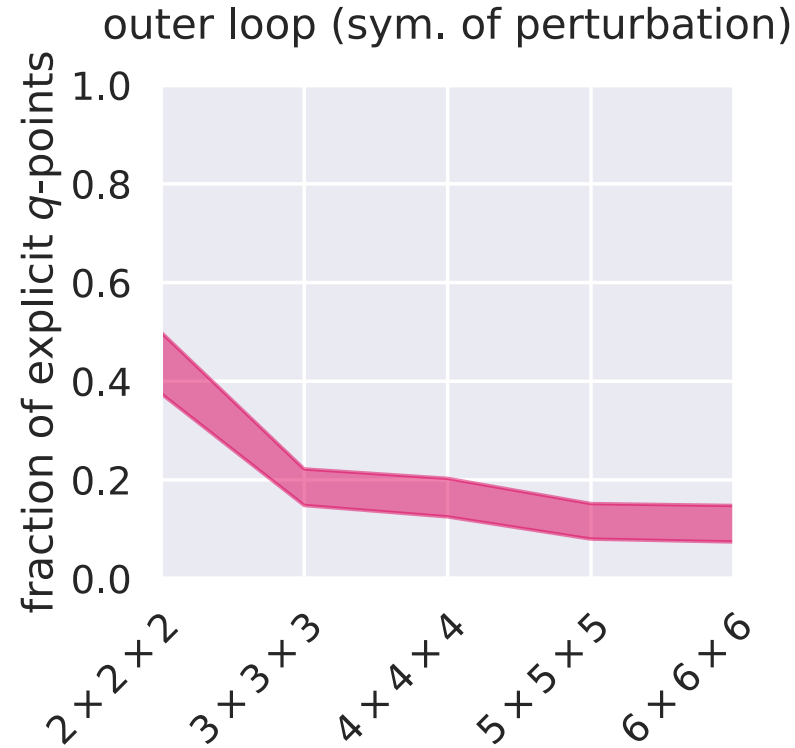
speedup of $\mathcal{O}(10)$ to $\mathcal{O}(100)$

Taking advantage of symmetries

To compute screening parameters via DFPT...

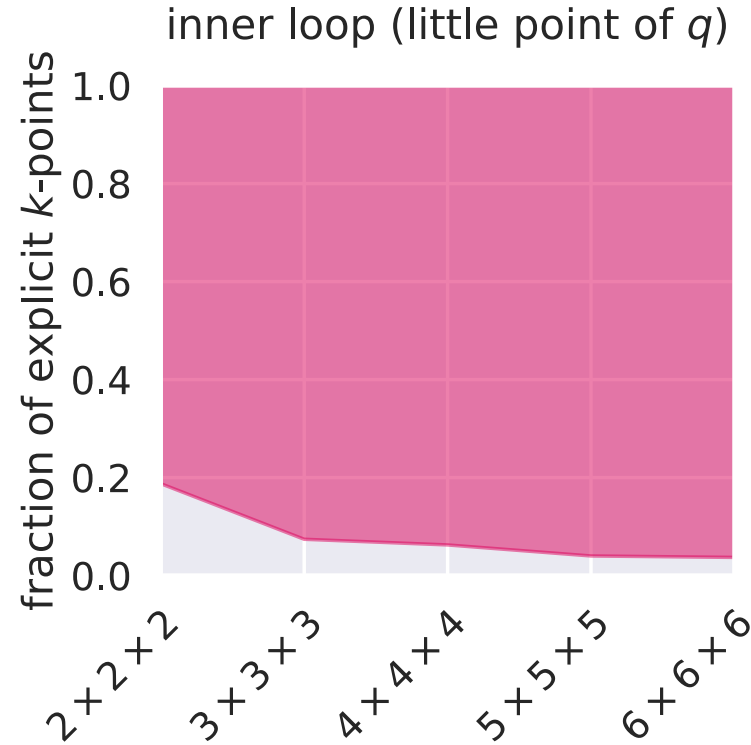
```
1: function CALCULATEALPHA( $n$ )
2:   for  $q \in \text{BZ}$  do
3:     for  $k \in \text{BZ}$  do
4:        $\triangleright$  Linear system  $Ax = b$  to obtain  $\Delta\psi_{\mathbf{k}+\mathbf{q},v}(\mathbf{r})$ 
5:     end
6:      $\Delta\rho_q^{0n} \leftarrow \sum_{\mathbf{k}v} \psi_{\mathbf{k}v}^*(\mathbf{r}) \Delta\psi_{\mathbf{k}+\mathbf{q},v}(\mathbf{r}) + c.c.$ 
7:      $\Pi_{0n,q}^{(r)} \leftarrow \langle \Delta\rho_q^{0n} | f_{\text{Hxc}} | \rho_q^{0n} \rangle$ 
8:      $\Pi_{0n,q}^{(u)} \leftarrow \langle \rho_q^{0n} | f_{\text{Hxc}} | \rho_q^{0n} \rangle$ 
9:   end
10:  return  $1 + \sum_q \Pi_{0n,q}^{(r)} / \sum_q \Pi_{0n,q}^{(u)}$ 
11:end
```

Taking advantage of symmetries



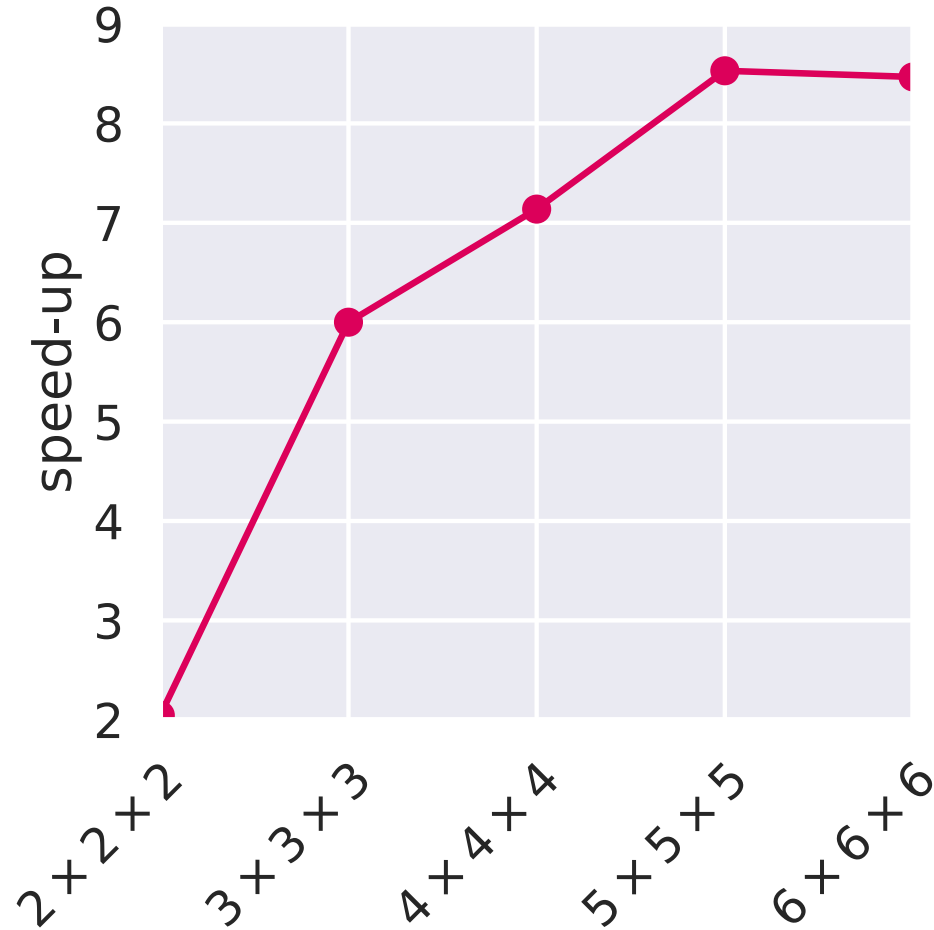
$\mathbf{q} \in \text{BZ} \rightarrow \mathbf{q} \in \text{IBZ}(n)$ (the symmetry of the perturbation; lower than that of the primitive cell)

Taking advantage of symmetries



$k \in \text{BZ} \rightarrow k \in \text{IBZ}(q)$ (can only use symmetries that leave q invariant)

Taking advantage of symmetries



Running a Koopmans functional calculation

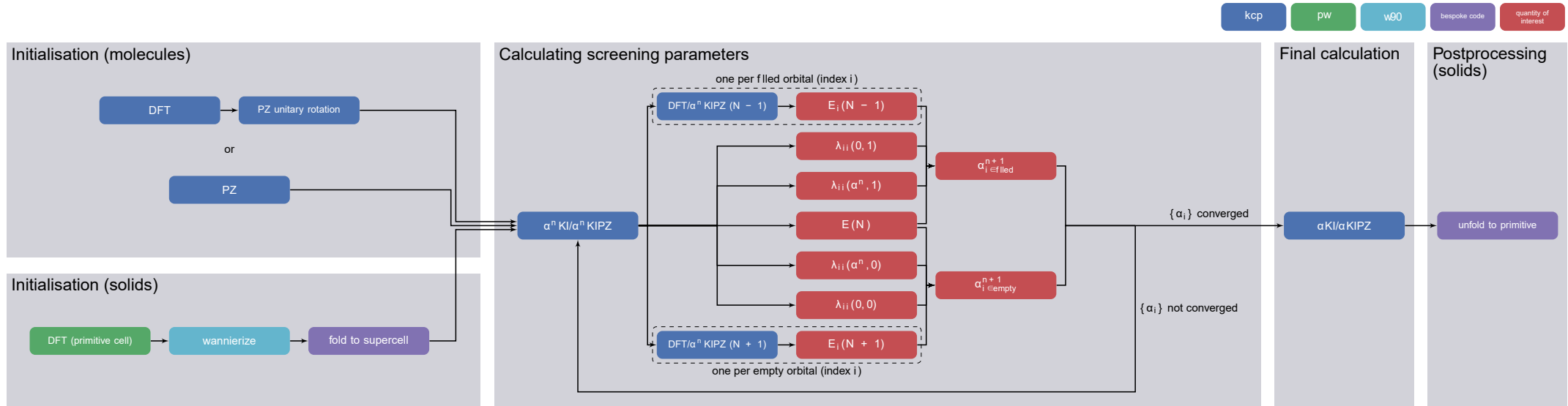
The workflows

The general workflow:

- define/initialize a set of variational orbitals
- calculate the screening parameters $\{\alpha_i\}$
- construct and diagonalize the Hamiltonian

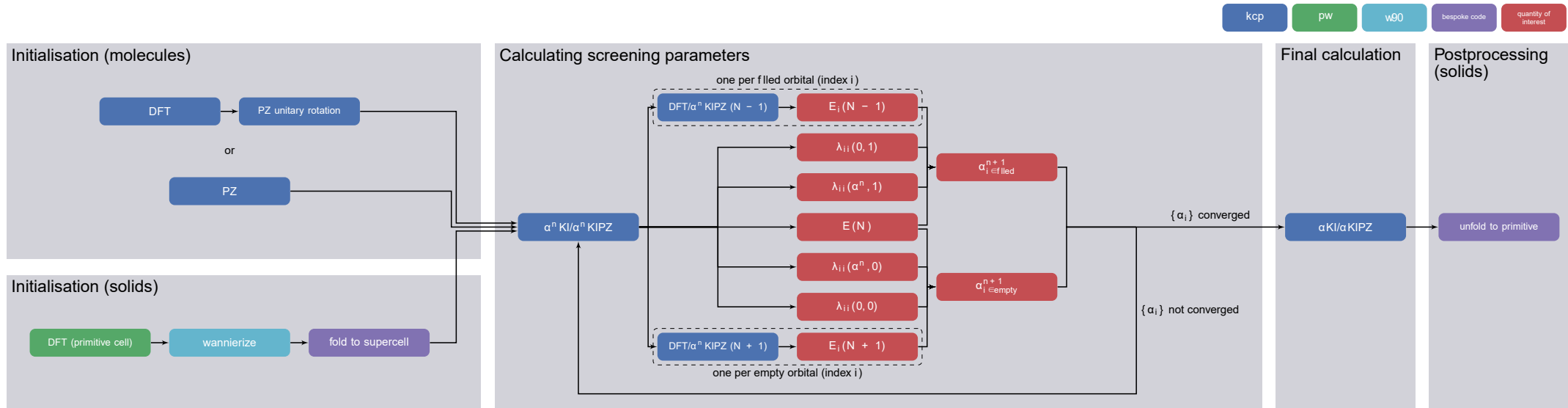
The workflows

(a) finite difference calculations using a supercell

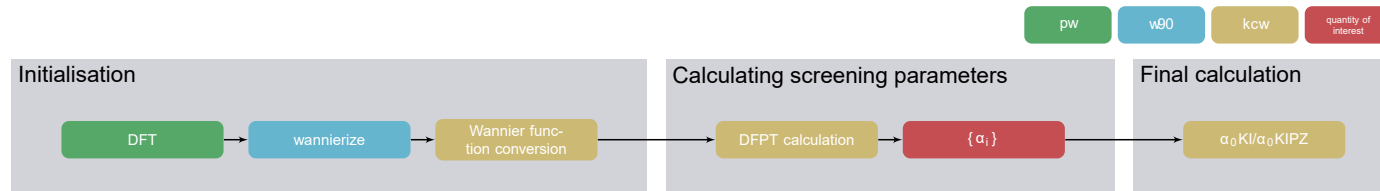


The workflows

(a) finite difference calculations using a supercell



(b) DFPT in a primitive cell



How do I run these calculations?

Complicated workflows mean that...

How do I run these calculations?

Complicated workflows mean that...

- lots of different codes that need to handshake

How do I run these calculations?

Complicated workflows mean that...

- lots of different codes that need to handshake
- lots of scope for human error

How do I run these calculations?

Complicated workflows mean that...

- lots of different codes that need to handshake
- lots of scope for human error
- reproducibility becomes difficult

How do I run these calculations?

Complicated workflows mean that...

- lots of different codes that need to handshake
- lots of scope for human error
- reproducibility becomes difficult
- expert knowledge required

How do I run these calculations?

Complicated workflows mean that...

- lots of different codes that need to handshake
- lots of scope for human error
- reproducibility becomes difficult
- expert knowledge required

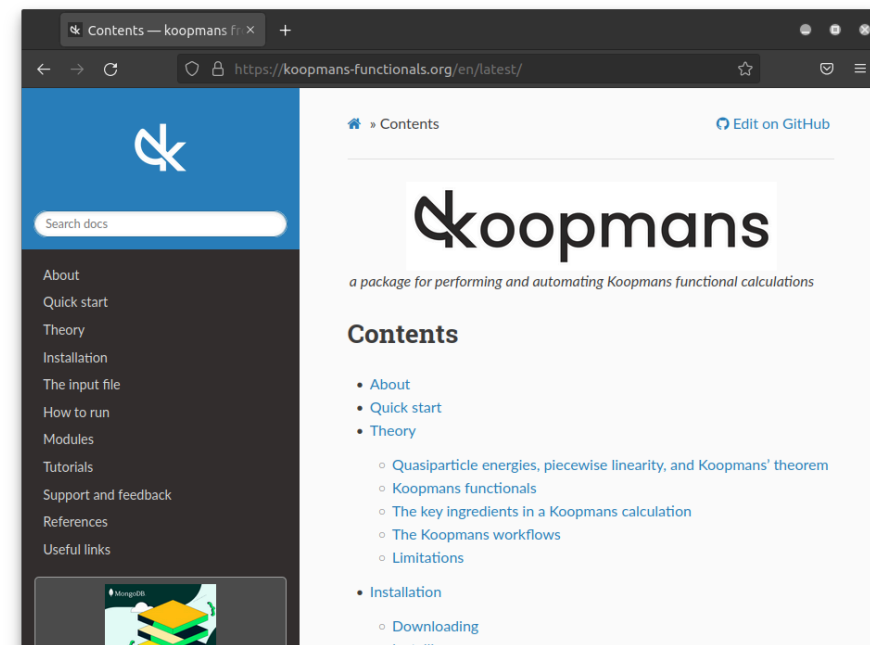
Our solution...

koopmans

The koopmans package

koopmans-functionals.org

- v1.0 released in 2023 (now v1.2)
- implementations of Koopmans functionals
- automated workflows
 - start-to-finish Koopmans calculations
 - Wannierisation
 - dielectric tensor
 - ...
- built on top of ASE
- under the hood, calls QUANTUM ESPRESSO
- does not require expert knowledge



E. B. Linscott *et al.* *J. Chem. Theory Comput.* **19**, 7097–7111 (2023)

A. H. Larsen *et al.* *J. Phys. Condens. Matter* **29**, 273002 (2017)

koopmans: the input file



```
{
  "workflow": {
    "task": "singlepoint",
    "functional": "ki",
    "method": "dscf",
    "init_orbitals": "mlwfs",
    "alpha_guess": 0.1
  },
  "atoms": {
    "atomic_positions": {
      "units": "crystal",
      "positions": [
        ["Si", 0.00, 0.00, 0.00],
        ["Si", 0.25, 0.25, 0.25]
      ]
    },
    "cell_parameters": {
      "periodic": true,
      "ibrav": 2,
      "celldm(1)": 10.262
    }
  },
}
```

```
  "k_points": {
    "grid": [8, 8, 8],
    "path": "LGXKG"
  },
  "calculator_parameters": {
    "ecutwfc": 60.0,
    "w90": {
      "projections": [
        [{"fsite": [0.125, 0.125, 0.125],
          "ang_mtm": "sp3"}],
        [{"fsite": [0.125, 0.125, 0.125],
          "ang_mtm": "sp3"}]
      ],
      "dis_froz_max": 11.5,
      "dis_win_max": 17.0
    }
  }
}
```

koopmans is scriptable



```
from ase.build import bulk
from koopmans.kpoints import Kpoints
from koopmans.projections import ProjectionBlocks
from koopmans.workflows import SinglepointWorkflow

# Use ASE to create bulk silicon
atoms = bulk('Si')

# Define the projections for the Wannierization (same for filled and empty manifold)
si_proj = [{'fsite': [0.25, 0.25, 0.25], 'ang_mtm': 'sp3'}]
si_projs = ProjectionBlocks.from_list([si_proj, si_proj], atoms=atoms)

# Create the workflow
workflow = SinglepointWorkflow(atoms = atoms,
                               projections = si_projs,
                               ecutwfc = 40.0,
                               kpoints = Kpoints(grid=[8, 8, 8], path='LGXKG', cell=atoms.cell),
                               calculator_parameters = {'pw': {'nbnd': 10},
                                                         'w90': {'dis_froz_max': 10.6, 'dis_win_max': 16.9}})

# Run the workflow
workflow.run()
```

koopmans is scriptable



```
from ase.build import bulk
from koopmans.kpoints import Kpoints
from koopmans.projections import ProjectionBlocks
from koopmans.workflows import SinglepointWorkflow

# Use ASE to create bulk silicon
atoms = bulk('Si')

# Define the projections for the Wannierization (same for filled and empty manifold)
si_proj = [{'fsite': [0.25, 0.25, 0.25], 'ang_mtm': 'sp3'}]
si_projs = ProjectionBlocks.from_list([si_proj, si_proj], atoms=atoms)

# Create the workflow
workflow = SinglepointWorkflow(atoms = atoms,
                               projections = si_projs,
                               ecutwfc = 40.0,
                               kpoints = Kpoints(grid=[8, 8, 8], path='LGXKG', cell=atoms.cell),
                               calculator_parameters = {'pw': {'nbnd': 10},
                                                         'w90': {'dis_froz_max': 10.6, 'dis_win_max': 16.9}})

# Run the workflow
workflow.run()
```

but don't get too used to it...

koopmans is scriptable



```
from ase.build import bulk
from koopmans.kpoints import Kpoints
from koopmans.projections import ProjectionBlocks
from koopmans.workflows import SinglepointWorkflow

# Use ASE to create bulk silicon
atoms = bulk('Si')

# Define the projections for the Wannierization (same for filled and empty manifold)
si_proj = [{'fsite': [0.25, 0.25, 0.25], 'ang_mtm': 'sp3'}]
si_projs = ProjectionBlocks.from_list([si_proj, si_proj], atoms=atoms)

# Create the workflow
workflow = SinglepointWorkflow(atoms = atoms,
                               projections = si_projs,
                               ecutwfc = 40.0,
                               kpoints = Kpoints(grid=[8, 8, 8], path='LGXKG', cell=atoms.cell),
                               calculator_parameters = {'pw': {'nbnd': 10},
                                                         'w90': {'dis_froz_max': 10.6, 'dis_win_max': 16.9}})

# Run the workflow
workflow.run()
```

but don't get too used to it...

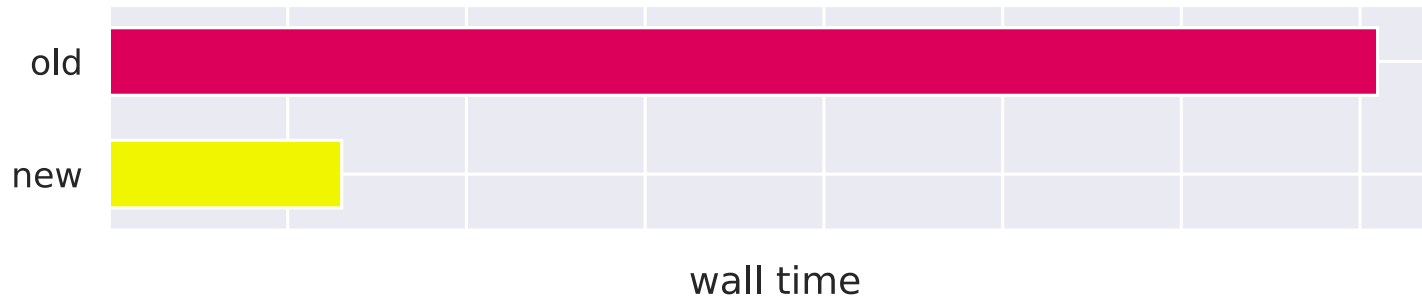
 **koopmans v2 is coming...** 

Koopmans AiiDA

- remote compute
- parallel step execution
- provenance-tracking
- error recovery
- and more!

Koopmans AiiDA

- remote compute
- parallel step execution
- provenance-tracking
- error recovery
- and more!



Automated Wannierisation

Koopmans functionals rely heavily on Wannier functions...

- to initialise the minimising orbitals, *or*
- in place of the minimising orbitals entirely

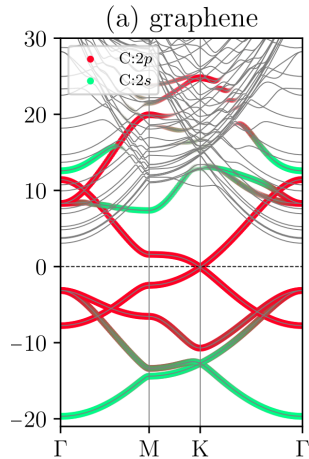
J. Qiao *et al. npj Comput Mater* **9**, 208 (2023)

J. Qiao *et al. npj Comput Mater* **9**, 206 (2023)

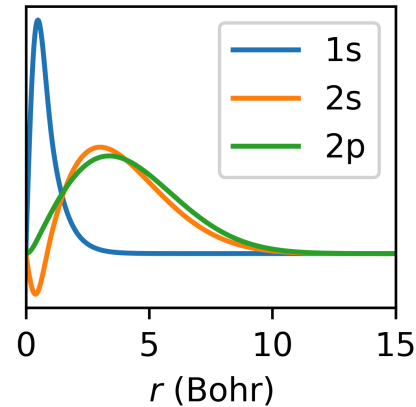
Automated Wannierisation

Koopmans functionals rely heavily on Wannier functions...

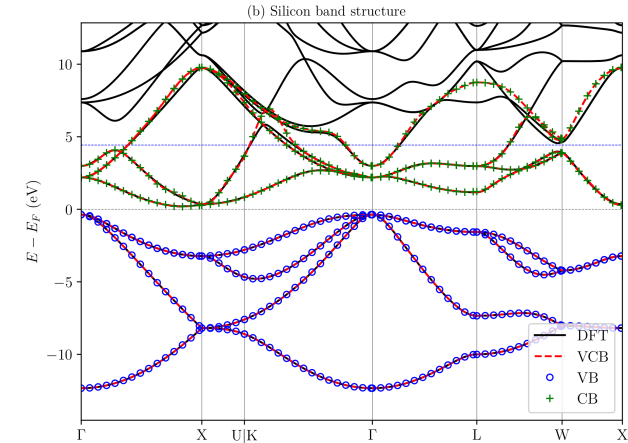
- to initialise the minimising orbitals, *or*
- in place of the minimising orbitals entirely



projectability-based
disentanglement

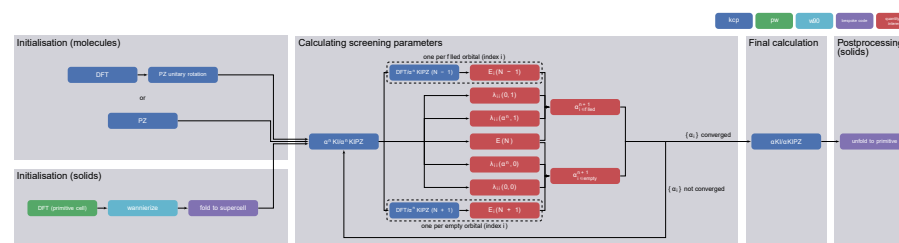
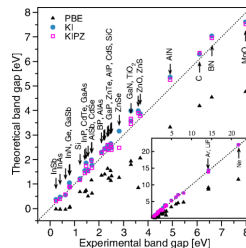
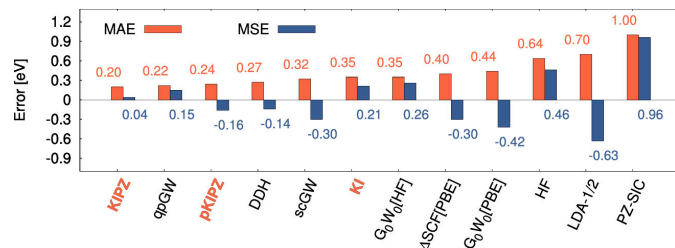


use PAOs found in
pseudopotentials



parallel transport to separate manifolds

Take home messages



- Koopmans functionals are more complicated than a simple semi-local DFT calculation, because of...
 - orbital-density-dependence
 - screening parameters
- Koopmans functionals are implemented in QUANTUM ESPRESSO
- the complexity of the workflows are handled by the koopmans package
- keep your eyes peeled for v2!

koopmans: An Open-Source Package for Accurately and Efficiently Predicting Spectral Properties with Koopmans Functionals

Edward B. Linscott,^{*,△} Nicola Colonna,[△] Riccardo De Gennaro, Ngoc Linh Nguyen, Giovanni Borghi, Andrea Ferretti, Ismaila Dabo, and Nicola Marzari^{*}



Cite This: <https://doi.org/10.1021/acs.jctc.3c00652>



Read Online

ACCESS |



Metrics & More

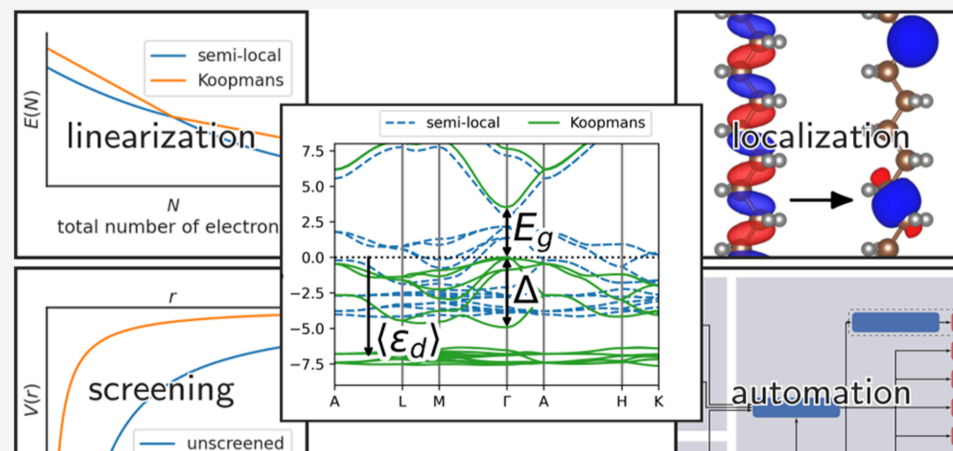


Article Recommendations



Supporting Information

ABSTRACT: Over the past decade we have developed Koopmans functionals, a computationally efficient approach for predicting spectral properties with an orbital-density-dependent functional framework. These functionals impose a generalized piecewise linearity condition to the entire electronic manifold, ensuring that orbital energies match the corresponding electron removal/addition energy differences (in contrast to semilocal DFT, where a mismatch between the two lies at the heart of the band gap problem and, more generally, the unreliability of Kohn–Sham orbital energies). This strategy has proven to be very powerful,



Thank you!

Acknowledgements



Nicola
Colonna



Miki Bonacci



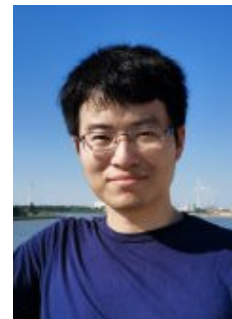
Aleksandr
Poliukhin



Marija
Stojkovic



Giovanni
Cistaro



Junfeng
Qiao



Yannick
Schubert



Nicola
Marzari



**Swiss National
Science Foundation**



Want to find out more? Go to koopmans-functionals.org

slides available at  [github/elinscott-talks](https://github.com/elinscott-talks)

- I. Dabo *et al.* Koopmans' Condition for Density-Functional Theory. *Phys. Rev. B* **82**, 115121 (2010).
- G. Borghi, A. Ferretti, N. L. Nguyen, I. Dabo & N. Marzari. Koopmans-Compliant Functionals and Their Performance against Reference Molecular Data. *Phys. Rev. B* **90**, 75135 (2014).
- N. Colonna, N. L. Nguyen, A. Ferretti & N. Marzari. Koopmans-Compliant Functionals and Potentials and Their Application to the GW100 Test Set. *J. Chem. Theory Comput.* **15**, 1905–1914 (2019).
- G. Borghi, C. H. Park, N. L. Nguyen, A. Ferretti & N. Marzari. Variational Minimization of Orbital-Density-Dependent Functionals. *Phys. Rev. B* **91**, 155112 (2015).
- N. L. Nguyen, N. Colonna, A. Ferretti & N. Marzari. Koopmans-Compliant Spectral Functionals for Extended Systems. *Phys. Rev. X* **8**, 21051 (2018).
- A. Ferretti, I. Dabo, M. Cococcioni & N. Marzari. Bridging Density-Functional and Many-Body Perturbation Theory: Orbital-Density Dependence in Electronic-Structure Functionals. *Phys. Rev. B* **89**, 195134 (2014).
- C. Li & W. Yang. On the Piecewise Convex or Concave Nature of Ground State Energy as a Function of Fractional Number of Electrons for Approximate Density Functionals. *J. Chem. Phys.* **146**, 74107 (2017).
- N. Colonna, R. De Gennaro, E. Linscott & N. Marzari. Koopmans Spectral Functionals in Periodic Boundary Conditions. *J. Chem. Theory Comput.* **18**, 5435–5448 (2022).
- R. De Gennaro, N. Colonna, E. Linscott & N. Marzari. Bloch's Theorem in Orbital-Density-Dependent Functionals: Band Structures from Koopmans Spectral Functionals. *Phys. Rev. B* **106**, 35106 (2022).
- N. Colonna, N. L. Nguyen, A. Ferretti & N. Marzari. Screening in Orbital-Density-Dependent Functionals. *J. Chem. Theory Comput.* **14**, 2549–2557 (2018).
- N. L. Nguyen, G. Borghi, A. Ferretti, I. Dabo & N. Marzari. First-Principles Photoemission Spectroscopy and Orbital Tomography in Molecules from Koopmans-Compliant Functionals. *Phys. Rev. Lett.* **114**, 166405 (2015).
- M. Stojkovic, E. B. Linscott & N. Marzari. Predicting the suitability of photocatalysts for water splitting using Koopmans spectral functionals: The case of TiO₂ polymorphs. *Phys. Rev. Mater.* **10**, 25403 (2026).
- Y. Schubert, N. Marzari & E. Linscott. Testing Koopmans Spectral Functionals on the Analytically Solvable Hooke's Atom. *J. Chem. Phys.* **158**, 144113 (2023).
- N. Q. Su, A. Mahler & W. Yang. Preserving Symmetry and Degeneracy in the Localized Orbital Scaling Correction Approach. *J. Phys. Chem. Lett.* **11**, 1528–1535 (2020).
- V. I. Anisimov & A. V. Kozhevnikov. Transition State Method and Wannier Functions. *Phys. Rev. B* **72**, 75125 (2005).
- L. Kronik, T. Stein, S. Refaely-Abramson & R. Baer. Excitation Gaps of Finite-Sized Systems from Optimally Tuned Range-Separated Hybrid Functionals. *J. Chem. Theory Comput.* **8**, 1515–1531 (2012).

- D. Wing *et al.* Band Gaps of Crystalline Solids from Wannier-localization–Based Optimal Tuning of a Screened Range-Separated Hybrid Functional. *Proc. Natl. Acad. Sci.* **118**, e2104556118 (2021).
- E. Kraisler & L. Kronik. Piecewise Linearity of Approximate Density Functionals Revisited: Implications for Frontier Orbital Energies. *Phys. Rev. Lett.* **110**, 126403 (2013).
- J. Ma & L.-W. Wang. Using Wannier Functions to Improve Solid Band Gap Predictions in Density Functional Theory. *Sci. Rep.* **6**, 24924 (2016).
- J. H. Skone, M. Govoni & G. Galli. Nonempirical Range-Separated Hybrid Functionals for Solids and Molecules. *Phys. Rev. B* **93**, 235106 (2016).
- C. Li, X. Zheng, N. Q. Su & W. Yang. Localized Orbital Scaling Correction for Systematic Elimination of Delocalization Error in Density Functional Approximations. *Natl. Sci. Rev.* **5**, 203–215 (2018).
- R. De Gennaro, N. Colonna, E. Linscott & N. Marzari. Bloch's theorem in orbital-density-dependent functionals: Band structures from Koopmans spectral functionals. *Phys. Rev. B* **106**, 35106 (2022).
- Y. Schubert, S. Luber, N. Marzari & E. Linscott. Predicting electronic screening for fast Koopmans spectral functional calculations. *npj Comput Mater* **10**, 1–12 (2024).
- E. B. Linscott *et al.* koopmans: an open-source package for accurately and efficiently predicting spectral properties with Koopmans Functionals. *J. Chem. Theory Comput.* **19**, 7097–7111 (2023).
- A. H. Larsen *et al.* The Atomic Simulation Environment—a Python Library for Working with Atoms. *J. Phys. Condens. Matter* **29**, 273002 (2017).
- S. P. Huber *et al.* AiiDA 1.0, a scalable computational infrastructure for automated reproducible workflows and data provenance. *Sci Data* **7**, 300 (2020).
- J. Qiao, G. Pizzi & N. Marzari. Projectability disentanglement for accurate and automated electronic-structure Hamiltonians. *npj Comput Mater* **9**, 208 (2023).
- J. Qiao, G. Pizzi & N. Marzari. Automated mixing of maximally localized Wannier functions into target manifolds. *npj Comput Mater* **9**, 206 (2023).

Spare slides

Recap from earlier

Key idea: construct a functional such that the *variational* orbital energies

$$\varepsilon_i^{\text{Koopmans}} = \langle \varphi_i | H | \varphi_i \rangle = \partial E_{\text{Koopmans}} / \partial f_i$$

are...

- independent of the corresponding occupancies f_i
- equal to the corresponding total energy difference $E_i(N-1) - E(N)$

zero band gap $\rightarrow$ occupancy matrix for variational orbitals is off-diagonal